

AWS FINAL EXAM CONTENT

AWS HANDS-ON PRACTICE & NETWORK SECURITY



DESIGN & DEPLOY HIGHLY AVAILABLE WEB ARCHITECTURE ON AWS

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NETWORKING

AWS Final Exam Content

This document provides an overview of the material related to your final exam. Use it as a reference to support your study.

Note: This document may have some errors. Please refer to the **official PPT** as your primary study resource.

This document is open source, created by Hussain Ali

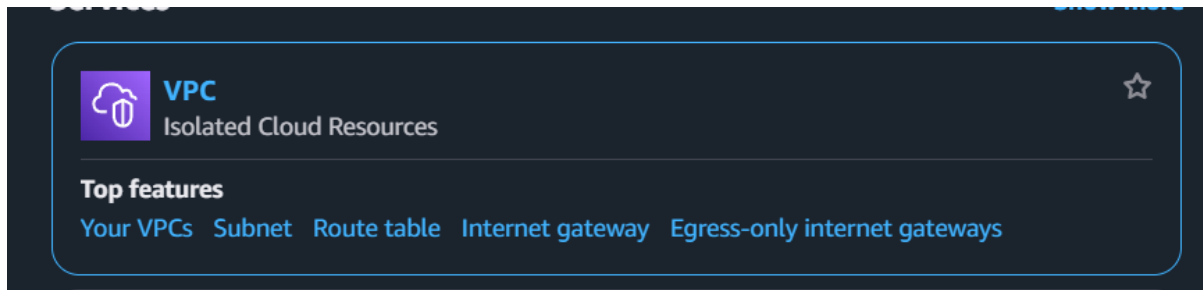
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How to Create a VPC

Create VPC



VPC dashboard

EC2 Global View

Filter by VPC:

Virtual private cloud

- Your VPCs**
- Subnets
- Route tables
- Internet gateways
- Egress-only Internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers [New](#)

Security

- Network ACLs
- Security groups

Your VPCs (1)

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set	Main route table
-	vpc-07f6a9bdc34923656	Available	Off	172.31.0.0/16	-	default-01f54125a5828ch...	rtb-0508bdf0dcff0a0b6

Create VPC

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create

☒ VPC only ☐ VPC and more

Name tag - optional

Creates a tag with a key of "Name" and a value that you specify.

my-vpc-01

IPv4 CIDR block

☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.0.0.0/24

CIDR block size must be between /16 and /28.

IPv6 CIDR block

☒ No IPv6 CIDR block ☐ IPAM-allocated IPv6 CIDR block ☐ Amazon-provided IPv6 CIDR block ☐ IPv6 CIDR owned by me

Tenancy

Default

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. No tags associated with the resource.

[Add tag](#)

You can add 50 more tags

[Cancel](#) [Preview code](#) [Create VPC](#)

Write the name of the VPC and the IPv4 CIDR (10.0.0.0/16), then choose **Create VPC**.

Create subnets

VPC dashboard

EC2 Global View

Filter by VPC:

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

Subnets (6)

Find subnets by attribute or tag

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 CIDR association
-	subnet-044e151627bb3d43d	Available	vpc-0f76a9bd63d923656	Off	172.31.48.0/20	-	-
-	subnet-0a654acfb2a006160	Available	vpc-0f76a9bd63d923656	Off	172.31.16.0/20	-	-
-	subnet-039dfa7032094a976	Available	vpc-0f76a9bd63d923656	Off	172.31.32.0/20	-	-
-	subnet-030fac9b13851c80	Available	vpc-0f76a9bd63d923656	Off	172.31.0.0/20	-	-
-	subnet-04bf9b1a28b074b93	Available	vpc-0f76a9bd63d923656	Off	172.31.64.0/20	-	-
-	subnet-06beea5d5e91af1a4	Available	vpc-0f76a9bd63d923656	Off	172.31.80.0/20	-	-

Create subnet

VPC

VPC ID

vpc-04cb5ca57f8c71f8a (Hussain-VPC)

Associated VPC CIDRs

IPv4 CIDRs

10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

my-subnet-01

The name can be up to 256 characters long.

Availability Zone

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

IPv4 VPC CIDR block

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.0.0/20

Tags - optional

No tags associated with the resource.

Add new tag

You can add 50 more tags.

Remove

Add new subnet

First, choose the VPC that you created before. Secondly, enter a name for your subnet. Thirdly, choose the Availability Zone. Lastly, enter the IPv4 subnet and click on create subnet.

Note: In subnets, you must create two separate subnets, one for the public and another for the private, and each subnet must have two different Availability Zones (public, private).

Create Route Table

VPC dashboard

EC2 Global View

Filter by VPC:

- Virtual private cloud
 - Your VPCs
 - Subnets
 - Route tables**
 - Internet gateways
 - Egress-only Internet gateways
 - Carrier gateways
 - DHCP option sets
 - Elastic IPs
 - Managed prefix lists
 - NAT gateways
 - Peering connections
 - Route servers
- Security
 - Network ACLs
 - Security groups

Route tables (2)

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
-	rtb-0508b0b0c0ff0a0b8	-	-	Yes	vpc-076a9bd65d923656	533267121457
-	rtb-0436e1bf5fb23a0fc	-	-	Yes	vpc-04cb5ca57f8c71f8a Hussa...	533267121457

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

my-route-table-01

VPC
The VPC to use for this route table.

vpc-04cb5ca57f8c71f8a (Hussain-VPC)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add 50 more tags.

Cancel Create route table

For the route table, first, choose a name. Second, select your VPC that you created before. Click on **Create route table**.

Note: You must create two route tables, one for the public and one for the private subnets.

Route tables (1/4)

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
-	rtb-0436e1bf5fb23a0fc	-	-	Yes	vpc-04cb5ca57f8c71f8a Hussa...	533267121457
-	rtb-0508b0b0c0ff0a0b8	-	-	Yes	vpc-076a9bd65d923656	533267121457
☒ Hussain-Public-RT	rtb-05978a7528869ff85	-	-	No	vpc-04cb5ca57f8c71f8a Hussa...	533267121457
☐ Hussain-Private-RT	rtb-09f404461fca786ab	-	-	No	vpc-04cb5ca57f8c71f8a Hussa...	533267121457

rtb-05978a7528869ff85 / Hussain-Public-RT

Details | Routes | **Subnet associations** | Edge associations | Route propagation | Tags

Explicit subnet associations (0)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
No subnet associations			

You do not have any subnet associations.

Edit subnet associations

After creating the route table, you must add the subnets. First, choose the public route table and from the drop-down menu select **Subnet associations**, then choose **Edit subnet associations**.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (3)

Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐ Hussain-private1	subnet-03fed7a668f6cc925	10.0.0.0/27	-	Main (rtb-0436e1bf5fb23a0fc)
☐ Hussain-private2	subnet-0908564e429afcfa	10.0.1.0/27	-	Main (rtb-0436e1bf5fb23a0fc)
☒ Hussain-Public	subnet-0d803b034e3c86695	10.0.2.0/28	-	Main (rtb-0436e1bf5fb23a0fc)

Cancel Save associations

Choose the public subnet, then click save associations.

Route tables (1/5) info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
-	rtb-0436e1bf5fb23a0fc	-	-	Yes	vpc-04cb5ca57f8c71f8a Hussa...	533267121457
-	rtb-0508bd0cd0a0b8	-	-	Yes	vpc-0f76a9bd63d923656	533267121457
Hussain-Public-RT	rtb-05978a7528869ff85	subnet-0d803b034e3c86...	-	No	vpc-04cb5ca57f8c71f8a Hussa...	533267121457
Hussain-Private-RT1	rtb-09f404461fca786ab	-	-	No	vpc-04cb5ca57f8c71f8a Hussa...	533267121457
Hussain-Private-RT2	rtb-0ad962d3612c9f585	-	-	No	vpc-04cb5ca57f8c71f8a Hussa...	533267121457

rtb-09f404461fca786ab / Hussain-Private-RT1

Details | Routes | **Subnet associations** | Edge associations | Route propagation | Tags

Explicit subnet associations (0)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
No subnet associations You do not have any subnet associations.			

[Edit subnet associations](#)

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/3)

Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
Hussain-private1	subnet-03fed7a668f6cc925	10.0.0.0/27	-	rtb-09f404461fca786ab / Hussain-Priv...
Hussain-private2	subnet-0908564e429afcfca	10.0.1.0/27	-	rtb-0ad962d3612c9f585 / Hussain-Pri...
Hussain-Public	subnet-0d803b034e3c86695	10.0.2.0/28	-	rtb-05978a7528869ff85 / Hussain-Pub...

Selected subnets

subnet-03fed7a668f6cc925 / Hussain-private1

[Cancel](#) [Save associations](#)

We do the same for the private subnet.

Create Internet Gateways

VPC dashboard <

EC2 Global View 

Filter by VPC: 

▼ **Virtual private cloud**

- Your VPCs
- Subnets
- Route tables
- Internet gateways**
- Egress-only Internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers [New](#)

▼ **Security**

- Network ACLs
- Security groups

Internet gateways (1) [Info](#)

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID	Owner
-	igw-07b2598c6bf4f7bc5	Attached	vpc-0f76a9bd63d923656	533267121457

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

my-internet-gateway

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.

[Add new tag](#)
You can add 50 more tags.

[Cancel](#) [Create internet gateway](#)

Enter a name for the internet gateway and click Create Internet Gateway.

Note: remember that the internet gateway is the one that will connect you with the internet; without it, you can't access the internet.

Internet gateways (2) [Info](#)

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID	Owner
-	igw-07b2598c6bf4f7bc5	Attached	vpc-0f76a9bd63d923656	533267121457
Hussain-IG	igw-0e3ae799808f80ec6	Detached	-	533267121457

After you create the internet gateway, you will notice that in the state it says that it is Detached; you have to attach it to your VPC.

Internet gateways (1/2) [Info](#)

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID	Owner
-	igw-07b2598c6bf4f7bc5	Attached	vpc-0f76a9bd63d923656	533267121457
☒ Hussain-IG	igw-0e3ae799808f80ec6	Detached	-	533267121457

[Actions](#) [Create internet gateway](#)

- View details
- Attach to VPC**
- Detach from VPC
- Manage tags
- Delete internet gateway

Attach to VPC (igw-0e3ae799808f80ec6) [Info](#)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

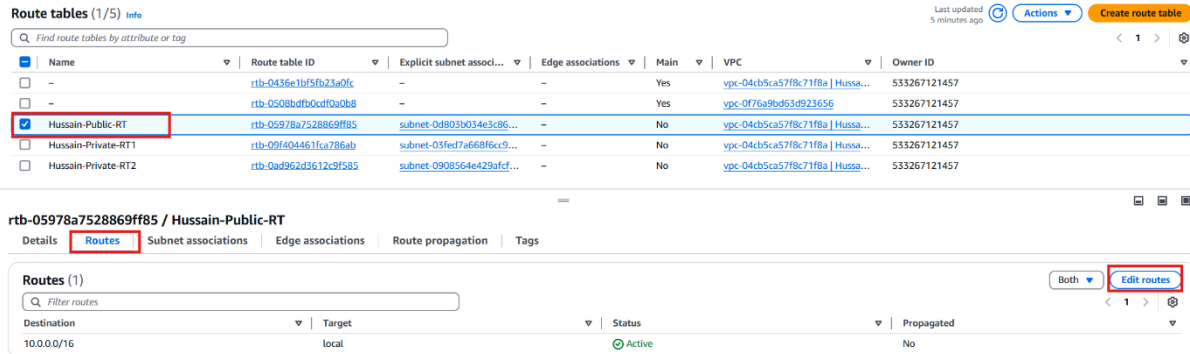
Select a VPC

vpc-04cb5ca57f8c71f8a - Hussain-VPC

[Cancel](#) [Attach internet gateway](#)

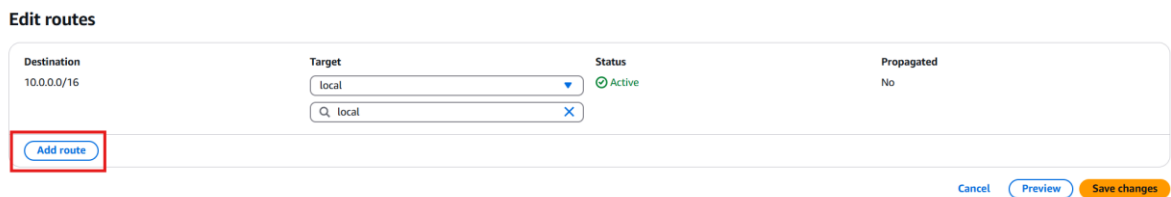
First, **select the internet gateway that you created**. Second, click on Actions, then **Attach to VPC**. Lastly, select your VPC and click on **Attach internet gateway**.

To connect the internet gateway with the public route table to have internet access:



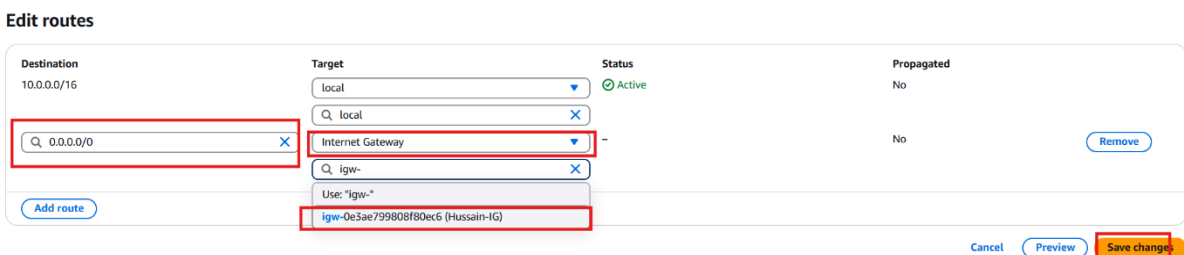
The screenshot shows the AWS Route Tables console. In the top table, 'Hussain-Public-RT' is selected. Below, the 'Routes' tab for 'rtb-05978a7528869ff85 / Hussain-Public-RT' is active. It shows a single route with destination '10.0.0.0/16' and target 'local'. The 'Edit routes' button is highlighted with a red box.

First, choose the public route, then from the drop-down menu choose routes, and then **Edit routes**.



The 'Edit routes' form is shown. The 'Destination' is '10.0.0.0/16', 'Target' is 'local', 'Status' is 'Active', and 'Propagated' is 'No'. The 'Add route' button is highlighted with a red box.

Click on **Add route**.

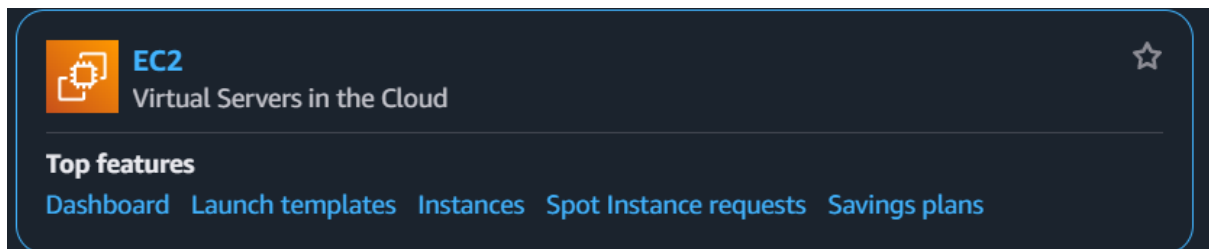


The 'Edit routes' form is shown. The 'Destination' is '0.0.0.0/0', 'Target' is 'Internet Gateway', 'Status' is 'Active', and 'Propagated' is 'No'. The 'Add route' button is highlighted with a red box.

First, choose 0.0.0.0/0, then for the target, choose internet gateway, and **select the internet gateway** that you created before. Finally, **click save changes**.

How to Create an EC2 instance

Create Ubuntu EC2



EC2

Dashboard

EC2 Global View 

Events

▼ Instances

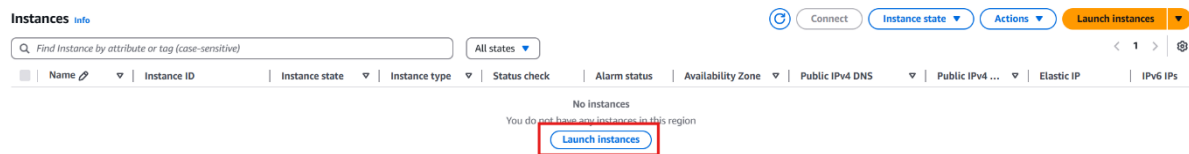
Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans



We will create a web server that runs Ubuntu.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Recents

My AMIs

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian


[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 AMI
ami-0e449927258d45bc4 (64-bit (x86), uefi-preferred) / ami-086a54924e40cab98 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

First, enter a name for the web server and choose Ubuntu.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

 [Create new key pair](#)

Scroll down to Key pair, choose Create new Key pair.

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

Cancel Create key pair

Choose a name for the key pair and click on **Create Key Pair**.

Network settings Info

Edit

Network | Info
vpc-Of76a9bd63d923656

Subnet | Info
No preference (Default subnet in any availability zone)

Auto-assign public IP | Info
Enable
Additional charges apply when outside of [free tier allowance](#)

Firewall (security groups) | Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow SSH traffic from
Helps you connect to your instance

Anywhere
0.0.0.0/0

☐ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

In network settings, click on **edit**.

▼ Network settings

Info

VPC - required

Info

vpc-0f76a9bd63d923656

(default)

172.31.0.0/16

Subnet

Info

No preference

Create new subnet

Auto-assign public IP

Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

launch-wizard-1

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and .-:/()#,@[]+=&:~!\$*

First, choose your VPC. Second, select the public subnet and make sure that the Auto-assign public IP is enabled. Last, **rename the Security group**.

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type

Info

ssh

▼

Protocol

Info

TCP

Port range

Info

22

Source type

Info

Anywhere

▼

Source

Info

🔍 Add CIDR, prefix list or security group

0.0.0.0/0 ✕

Description - optional

Info

e.g. SSH for admin desktop

⚠️ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

✕

Add security group rule

► Advanced network configuration

Scroll down in the **Inbound Security Group Rules**, click on **Add security group rule**.

▼ Security group rule 2 (All, All, 0.0.0.0/0) Remove

Type Info	Protocol Info	Port range Info
All traffic	All	All
Source type Info	Source Info	Description - optional Info
Custom	Add CIDR, prefix list or security group 0.0.0.0/0	e.g. SSH for admin desktop

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ×

Add security group rule

► Advanced network configuration

For the type, **choose All traffic**, and for the source, choose 0.0.0.0/0

Note: In the exam, you must choose the protocol that is given to you, and you can't choose all traffic.

▼ **Summary**

Number of instances Info

1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-084568db4383264d4

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

ℹ **Free tier:** In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. ×

Cancel Launch instance Preview code

Finally, click on **Launch instance**.

Instances (1/1) [Info](#)

Find instance by attribute or tag (case-sensitive) All states Last updated less than a minute ago Connect Instance state Actions Launch instances

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☒	webServer1	i-0c96dd20eca6f7f51	Running	t2.micro	Initializing	View alarms +	us-east-1b	-	52.91.158.104	-	-

There are two ways to connect to the web server: choosing your instance and clicking on Connect; the second way will be explained in the Red Hat EC2 instance.

Connect to instance [Info](#)

Connect to your instance i-0c96dd20eca6f7f51 (webServer1) using any of these options

EC2 Instance Connect | Session Manager | SSH client | EC2 serial console

Instance ID: i-0c96dd20eca6f7f51 (webServer1)

Connection Type

☒ Connect using EC2 Instance Connect
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 or IPv6 address.

☐ Connect using EC2 Instance Connect Endpoint
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

☒ Public IPv4 address
52.91.158.104

☐ IPv6 address
-

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.

ubuntu

Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel Connect

Choose **EC2 Instance Connect** and click on **Connect**.

```

AWS | Search | [Alt+S] | United States (N. Virginia) | verlabs/user2302147-i202202674@student.polytechnic.bh @ 5332-6...
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-1024-aws x86_64)
 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Apr 20 17:47:13 UTC 2025
System load: 0.16          Processes: 107
Usage of /: 25.0% of 6.71GB   Users logged in: 0
Memory usage: 22%          IPv4 address for enx0: 10.0.2.13
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

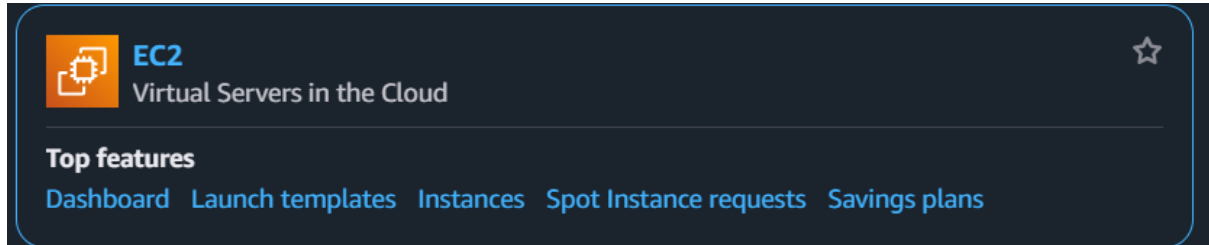
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-0-2-13:~$

```

If you have done everything correctly, it should open the terminal for you.

Creating Red Hat EC2:



EC2

Dashboard

EC2 Global View

Events

Instances

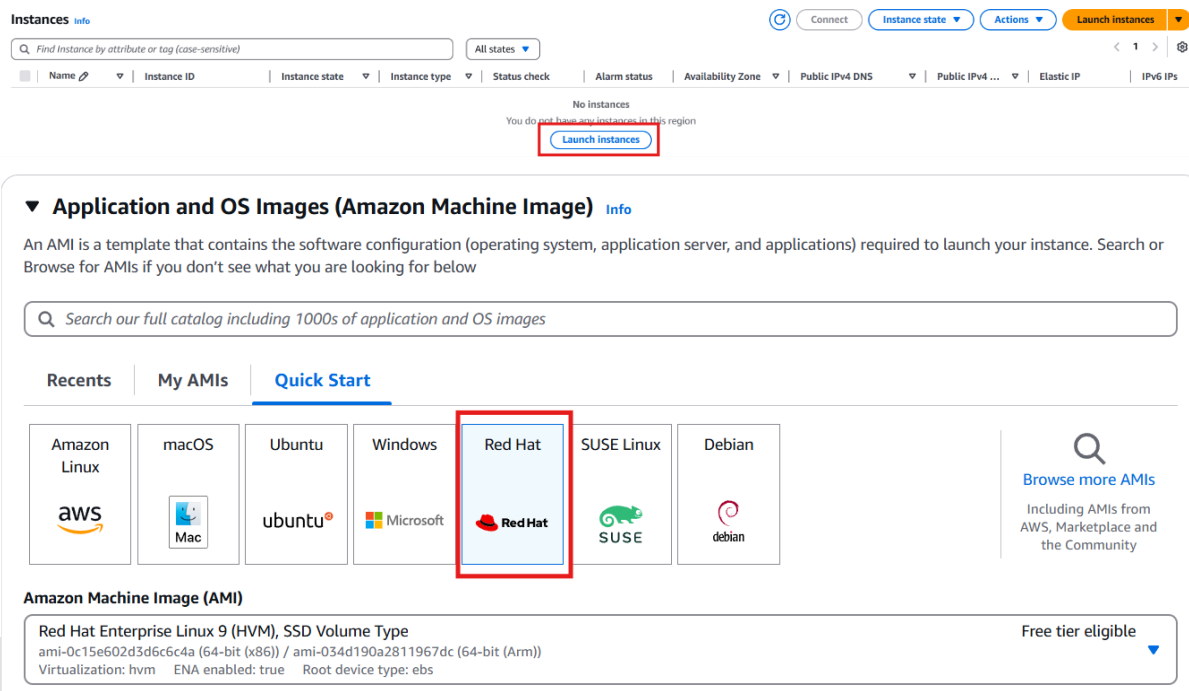
Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

A screenshot of the AWS Management Console showing the EC2 Instances page. The "Instances" tab is selected in the left sidebar. The main content area shows a message "No instances" and a "Launch instances" button. Below this, there is a section for "Application and OS Images (Amazon Machine Image)" with a search bar and a "Quick Start" tab. Under "Quick Start", the "Red Hat" option is highlighted with a red box. Below the "Quick Start" section, the "Amazon Machine Image (AMI)" section shows the "Red Hat Enterprise Linux 9 (HVM), SSD Volume Type" AMI, which is also highlighted with a red box. The AMI details include the ID "ami-0c15e602d3d6c6c4a (64-bit (x86)) / ami-034d190a2811967dc (64-bit (Arm))", virtualization type "hvm", ENA enabled "true", and root device type "ebs". A "Free tier eligible" badge is visible on the right.

Choose **Red Hat** from the **Quick Start** after entering the name of the EC2.

Scroll down to Key pair, choose **Create new Key pair**.

×

Key pair name
Key pairs allow you to connect to your instance securely.

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Cancel

Create key pair

Choose a name for the key pair and click on **Create Key Pair**.

▼ **Network settings** Info

Network | Info
vpc-0f76a9bd63d923656

Subnet | Info
No preference (Default subnet in any availability zone)

Auto-assign public IP | Info
Enable
Additional charges apply when outside of free tier allowance

Firewall (security groups) | Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow SSH traffic from
Helps you connect to your instance

Anywhere
0.0.0.0/0

☐ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Edit

In network settings, click on **edit**.

18

▼ Network settings

Info

VPC - required | Info

vpc-0f76a9bd63d923656

172.31.0.0/16

(default) ▼

↻

Subnet | Info

No preference ▼

↻ Create new subnet

Auto-assign public IP | Info

Enable ▼

Additional charges apply when outside of free tier allowance

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

launch-wizard-1

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and . - / () # . @ [] + = & ; ' ! \$ *

First, choose your VPC. Second, select the public subnet and **make sure** that the **Auto-assign public IP is enabled (Enable)**. Last, **rename the Security group**.

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type

Info

ssh

▼

Source type

Info

Anywhere

▼

Protocol

Info

TCP

Port range

Info

22

Source

Info

🔍 Add CIDR, prefix list or security group

0.0.0.0/0 ✕

Description - optional

Info

e.g. SSH for admin desktop

Remove

⚠️ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

✕

Add security group rule

► Advanced network configuration

Scroll down in the **Inbound Security Group Rules**, and click on **Add security group rule**.

▼ Security group rule 2 (All, All, 0.0.0.0/0) Remove

Type Info All traffic ▼	Protocol Info All	Port range Info All
Source type Info Custom ▼	Source Info Add CIDR, prefix list or security group 0.0.0.0/0 ✕	Description - optional Info e.g. SSH for admin desktop

 Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ✕

Add security group rule

► Advanced network configuration

For the type, choose **All traffic**, and for the source, choose **0.0.0.0/0**

▼ **Summary**

Number of instances | [Info](#)

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-084568db4383264d4

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

 **Free tier:** In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. ✕

Cancel Launch instance [Preview code](#)

Finally, click on **Launch instance**.

Instances (1/1) [info](#)

Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[All states](#)

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☒	webServer1	i-0c96dd20eca67751	Running	t2.micro	Initializing	View alarms	us-east-1b	-	52.91.158.104	-	-

To connect to the web server, **choose your instance** and click on **Connect**.

Connect to instance [info](#)

Connect to your instance i-0c96dd20eca67751 (webserver2) using any of these options

EC2 Instance Connect | Session Manager | **SSH client** | EC2 serial console

Instance ID
i-0c96dd20eca67751 (webserver2)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is h-key.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
`chmod 400 "h-key.pem"`
4. Connect to your instance using its Public IP:
54.87.123.119

Example:
`ssh -i "h-key.pem" ec2-user@54.87.123.119`

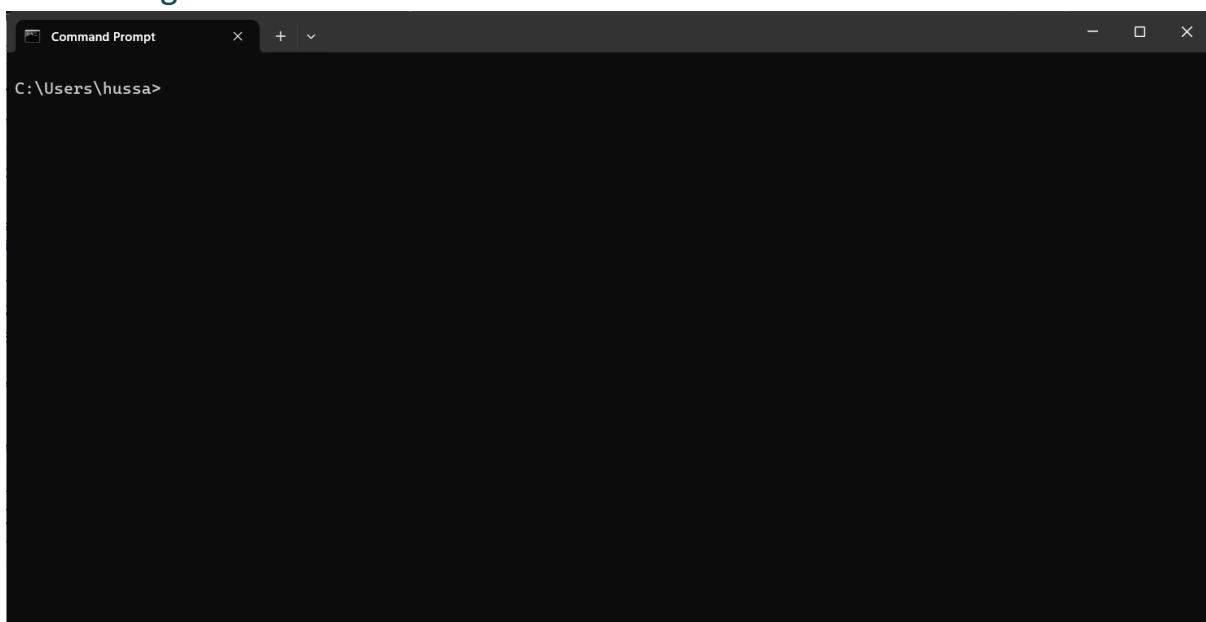
Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

[Cancel](#)

Choose **SSH client** and copy the **Example**.

Note: Red Hat Linux, you can't connect to it directly like Ubuntu; you have to connect to it from the CMD.

Connecting an EC2 instance from the CMD



Open the CMD and go to where the key you created was downloaded, and enter the command that you copied.

```
Command Prompt - ssh -i "h-key.pem" ec2-user@54.87.123.119
C:\Users\hussa\Downloads>ssh -i "h-key.pem" ec2-user@54.87.123.119
The authenticity of host '54.87.123.119 (54.87.123.119)' can't be established.
ED25519 key fingerprint is SHA256:pvaKZH1EHcd0YL8acVV7ugdYTF7+5yMWYkp5P7TpJ00.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

Enter **yes**.

```
ec2-user@ip-192-168-0-219:~
C:\Users\hussa\Downloads>ssh -i "h-key.pem" ec2-user@54.87.123.119
The authenticity of host '54.87.123.119 (54.87.123.119)' can't be established.
ED25519 key fingerprint is SHA256:pvaKZH1EHcd0YL8acVV7ugdYTF7+5yMWYkp5P7TpJ00.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.87.123.119' (ED25519) to the list of known hosts.
Register this system with Red Hat Insights: rhc connect

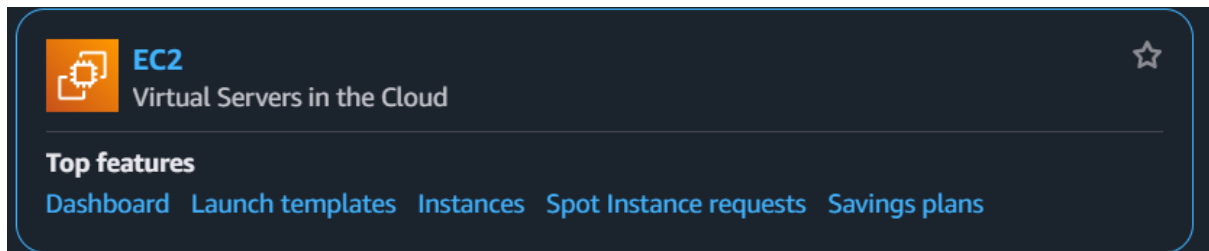
Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
[ec2-user@ip-192-168-0-219 ~]$ |
```

Now you will see that it is connected. Try running the command “**sudo yum update**” to update Red Hat.

Creating Windows EC2



EC2

Dashboard

EC2 Global View 

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Instances Info

All states

Name Instance ID Instance state Instance type Status check Alarm status Availability Zone Public IPv4 DNS Public IPv4 ... Elastic IP IPv6 IPs

No instances
You do not have any instances in this region

[Launch instances](#)

▼ **Application and OS Images (Amazon Machine Image)** Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Recents | **My AMIs** | **Quick Start**

Amazon Linux


macOS


Ubuntu


Windows


Red Hat


SUSE Linux


Debian


[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Microsoft Windows Server 2022 Base
ami-0c765d44cf1f25d26 (64-bit (x86))
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

Description
Microsoft Windows 2022 Datacenter edition. [English]
Microsoft Windows Server 2022 Full Locale English AMI provided by Amazon

Architecture	AMI ID	Publish Date	Username
64-bit (x86)	ami-0c765d44cf1f25d26	2025-03-12	Administrator

Verified provider

Choose Windows from the Quick start after entering the name for the EC2, and make sure that you choose **Microsoft Windows Server 2022 Base** from **Amazon Machine Image (AMI)**

Scroll down to **Key pair**, choose **Create new Key pair**.

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Cancel

Create key pair

Choose a name for the key pair and click on **Create Key Pair**.

Network settings

Info

Edit

Network

Info

vpc-0f76a9bd63d923656

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow SSH traffic from
Helps you connect to your instance

Anywhere
0.0.0.0/0

☐ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

In network settings, click on **edit**.

▼ Network settings

Info

VPC - required

Info

vpc-0f76a9bd63d923656

(default)

172.31.0.0/16

Subnet

Info

No preference

Create new subnet

Auto-assign public IP

Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

launch-wizard-1

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and .-:/()!#,@[]+=&:~|_\$*

First, choose your VPC. Second, select the public subnet and make sure that the Auto-assign public IP is enabled. Last, **rename the Security group**.

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type

Info

ssh

▼

Source type

Info

Anywhere

▼

Protocol

Info

TCP

Port range

Info

22

Source

Info

Q Add CIDR, prefix list or security group

0.0.0.0/0 X

Description - optional

Info

e.g. SSH for admin desktop

Remove

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

X

Add security group rule

► Advanced network configuration

Scroll down in the **Inbound Security Group Rules**, click on **Add security group rule**.

▼ Security group rule 2 (All, All, 0.0.0.0/0) Remove

Type Info	Protocol Info	Port range Info
All traffic	All	All
Source type Info	Source Info	Description - optional Info
Custom	Add CIDR, prefix list or security group 0.0.0.0/0	e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Add security group rule](#)

► Advanced network configuration

For the type, choose **All traffic**, and for the source, choose **0.0.0.0/0**

▼ Summary

Number of instances | Info

1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-084568db4383264d4

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

[Cancel](#) [Launch instance](#) [Preview code](#)

Finally, click on Launch instance.

Instances (1/1) [info](#)

Find instance by attribute or tag (case-sensitive) All states [▼](#)

Last updated less than a minute ago [Connect](#) [Instance state ▼](#) [Actions ▼](#) [Launch instances ▼](#)

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☒	webServer1	i-0c96dd20eca6f7f51	Running	t2.micro	Initializing	View alarms +	us-east-1b	-	52.91.158.104	-	-

To connect to the web server, **choose your instance** and click on **Connect**.

Connect to instance [info](#)

Connect to your instance i-0df849da29db676aa (WebServer3) using any of these options

Session Manager [RDP client](#) EC2 serial console

Instance ID [i-0df849da29db676aa \(WebServer3\)](#)

Connection Type

☒ Connect using RDP client
Download a file to use with your RDP client and retrieve your password.

☐ Connect using Fleet Manager
To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#).

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

[Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public IP [3.95.161.156](#)

Username [info](#)
[Administrator](#)

Password [Get password](#)

☐ If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

[Cancel](#)

First, go to the **RDP client** and click on **Get password**.

Get Windows password [info](#)

Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.

Instance ID [i-0df849da29db676aa \(WebServer3\)](#)

Key pair associated with this instance
[ihh](#)

Private key
Either upload your private key file or copy and paste its contents into the field below.

[Upload private key file](#)

Private key contents - optional

[Cancel](#) [Decrypt password](#)

Click on **Upload private key file** and upload the key you have created for the Windows EC2, then click on **Decrypt password**.

Connect to instance [Info](#)

Connect to your instance i-Odf849da29db676aa (WebServer3) using any of these options

Session Manager **RDP client** EC2 serial console

Instance ID
i-Odf849da29db676aa (WebServer3)

Connection Type

☒ **Connect using RDP client**
Download a file to use with your RDP client and retrieve your password.

☐ **Connect using Fleet Manager**
To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#)

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

[Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public IP
3.95.161.156

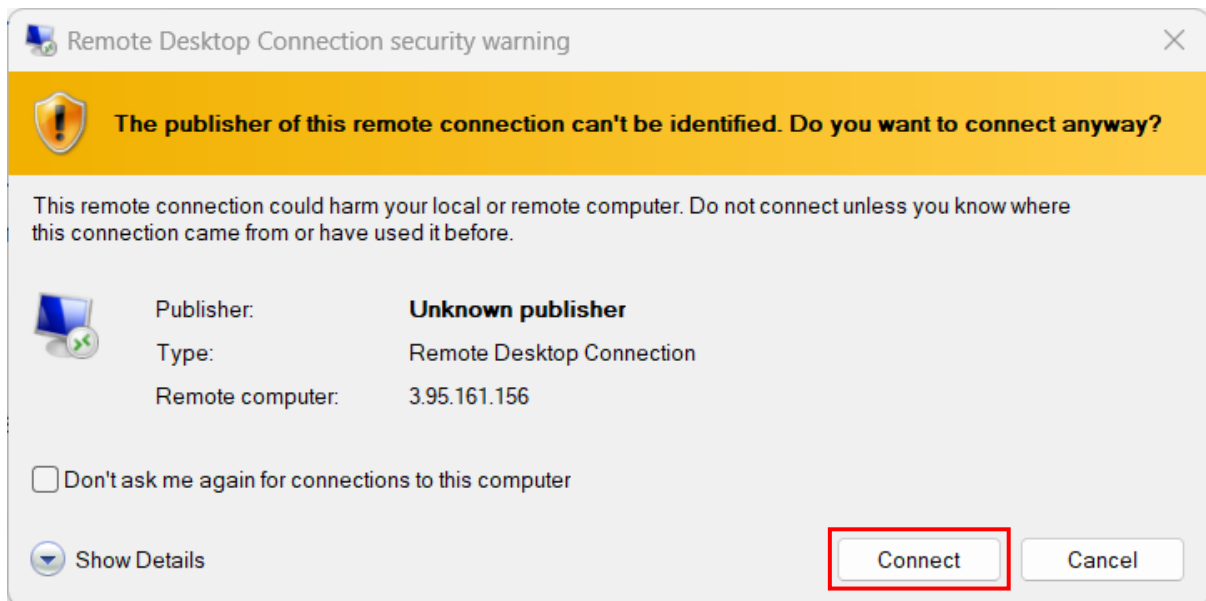
Username [Info](#)
Administrator

Password
Nx\$UcIr-LvJlthvZGv7EmTFNBi2&i-WS

☐ If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

[Cancel](#)

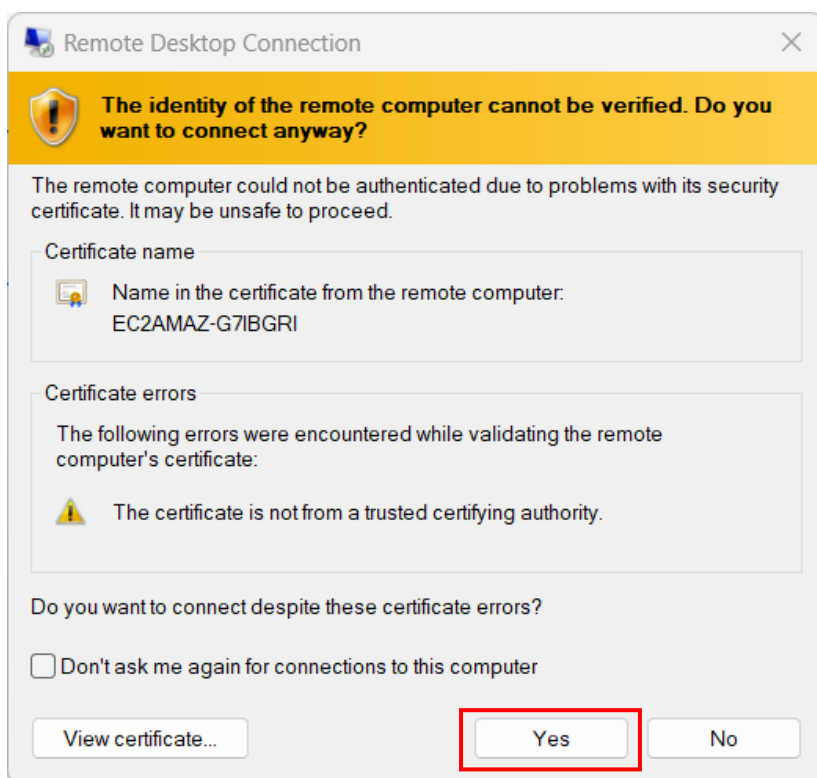
After decrypting the key, it will give you the password. Copy it. Then download the **Remote Desktop file** and open it.



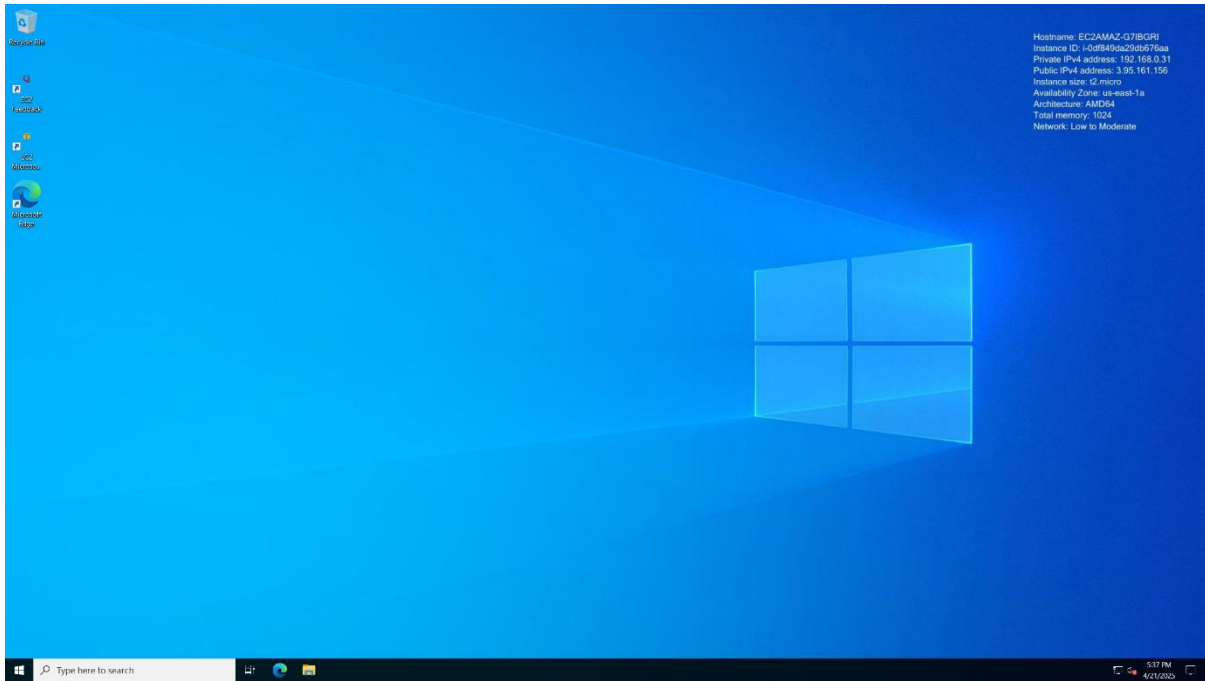
Click **connect**.



Paste the **password** that you copied before and click ok.



Choose **Yes**, and the window will start.

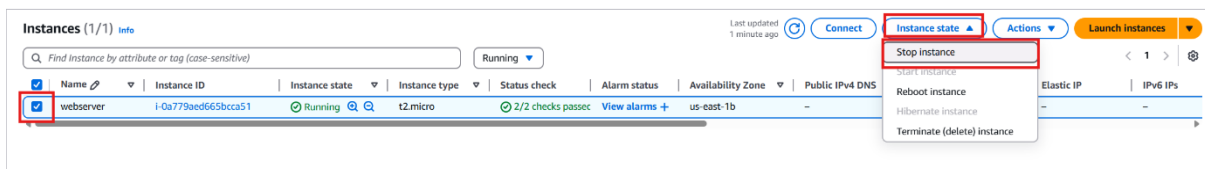
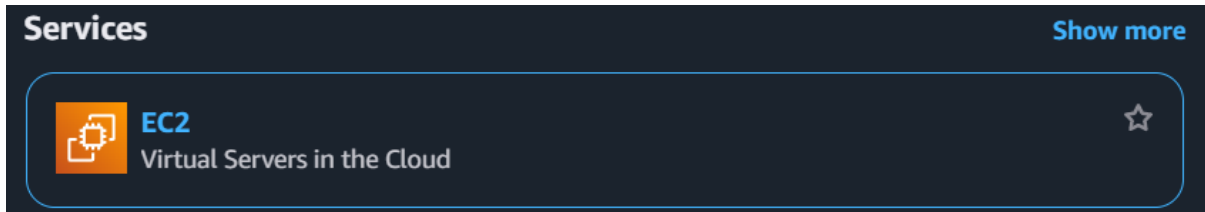


If you've done everything correctly, it will open for you like this.

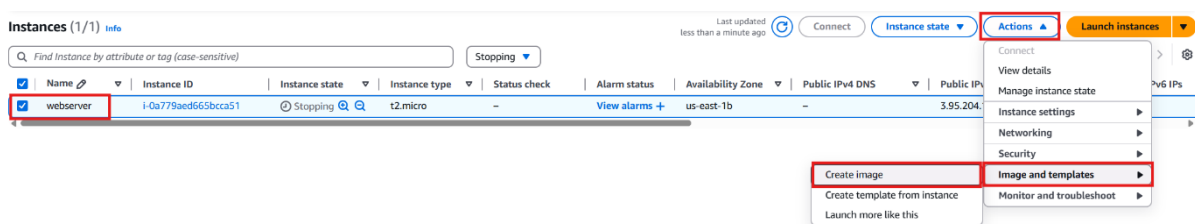
How to Create Load Balancers

Requirements: VPC with **2 subnets** (public and private in **different Availability Zones**) and an EC2 instance that runs Ubuntu with **Apache2** service on it, and an internet gateway.

Create an image (AMI) from EC2 instance



First, go to EC2. Second, select the instance and stop the web server that you created before from the instance state. It will show you in the Instance state **Stopped**.



Now you will **create an image** from the web server that you stopped. First, select your instance and click on **action** from the drop-down menu, select **image and templates**, then click on **create image**, which will create a copy of your original instance.

Create image [Info](#)

An image (also referred to as an AMI) defines the programs and settings that are applied when you launch an EC2 instance. You can create an image from the configuration of an existing instance.

Instance ID
i-0a779aed665bcca51 (webserver)

Image name

Enter image name

Maximum 127 characters. Can't be modified after creation.

Image description - optional

Image description

Maximum 255 characters

☒ Reboot instance

When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.

Instance volumes

Storage type	Device	Snapshot	Size	Volume type	IOPS	Throughput	Delete on termination	Encrypted
EBS	/dev/s...	Create new snapshot from v...	8	EBS General Purpose SSD - ...	3000		☒ Enable	☐ Enable

[Add volume](#)

During the image creation process, Amazon EC2 creates a snapshot of each of the above volumes.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

☒ Tag image and snapshots together

Tag the image and the snapshots with the same tag.

☐ Tag image and snapshots separately

Tag the image and the snapshots with different tags.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#)
[Create image](#)

Give your image a name and a meaningful description, and click **Create Image**.

EC2

Dashboard

EC2 Global View [↗](#)

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

▼ Images

AMIs

AMI Catalog

Amazon Machine Images (AMIs) (1) [Info](#)

Owned by me [Find AMI by attribute or tag](#)

☐	Name	AMI name	AMI ID	Source	Owner	Visibility	Status	Creation date
☐	Webserver2		ami-0dbbc2aeaba67a847	533267121457/Webserver2	533267121457	Private	Pending	2025/04/23 15:47 GMT+3

From the left-side menu, choose **AMIs**, and you will find the image you created there.

Note: if you have created the image, start the EC2 instance again, and stop it by taking the same steps for stopping the instance.

Instances (1/2) [Info](#)

Find instance by attribute or tag (case-sensitive) [All states](#)

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☐	webserver	i-0a779aed665bcca51	Stopped	t2.micro	2/2 checks passed	View alarms +	us-east-1b	-

Instance state [Actions](#) [Launch instances](#)

Stop instance

Start instance

Reboot instance

Hibernate instance

Terminate (delete) instance

Create a new public subnet



VPC

Isolated Cloud Resources

☆

VPC dashboard

EC2 Global View

Filter by VPC:

- ▼ Virtual private cloud
- Your VPCs
- Subnets**
- Route tables
- Internet gateways
- Egress-only Internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers New

Subnets (8)

Find subnets by attribute or tag

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 CIDR association IC
-	subnet-044e151627bb3d43d	Available	vpc-0f76a9bd63d923656	Off	172.31.48.0/20	-	-
-	subnet-0ac64acfb2a006160	Available	vpc-0f76a9bd63d923656	Off	172.31.16.0/20	-	-
-	subnet-039dfa7032094a976	Available	vpc-0f76a9bd63d923656	Off	172.31.32.0/20	-	-
-	subnet-030facf9b13851c80	Available	vpc-0f76a9bd63d923656	Off	172.31.0.0/20	-	-
-	subnet-04bf9b1a28b074b93	Available	vpc-0f76a9bd63d923656	Off	172.31.64.0/20	-	-
-	subnet-06b6ee5d5e91af1a4	Available	vpc-0f76a9bd63d923656	Off	172.31.80.0/20	-	-
Private	subnet-052475d556a2cc057	Available	vpc-0684c34a048585882 Hus...	Off	192.168.1.0/24	-	-
Public	subnet-0514d1738c5c31d2	Available	vpc-0684c34a048585882 Hus...	Off	192.168.2.0/24	-	-

Last updated 39 minutes ago

Actions Create subnet

You must create a new public subnet.

Create subnet

VPC

VPC ID

Create subnets in this VPC.

vpc-0684c34a048585882 (Hussain-VPC)

Associated VPC CIDRs

IPv4 CIDRs: 192.168.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

my-subnet-01

The name can be up to 256 characters long.

Availability Zone

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

IPv4 VPC CIDR block

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/16

IPv4 subnet CIDR block

192.168.0.0/20

▼ Tags - optional

No tags associated with the resource.

Add new tag

You can add 50 more tags.

Remove

Add new subnet

Cancel
Create subnet

now go to VPC under subnets. Click on **Create subnet**.

First, choose the VPC that you created before. Second, enter a name for your new public subnet. Third, choose a different available Zone from the other 2 subnets (public, private) that you created before. Finally, choose an IPV4 CIDR and click **Create Subnet**.

Create a Route table for the new subnet

VPC dashboard <

EC2 Global View 

Filter by VPC: 

▼ **Virtual private cloud**

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only Internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers **New**

Route tables (4) Info

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	-	rtb-0508bdf0c0ffa0b8	-	-	Yes	vpc-0f76a9bd63d923656	533267121457
☐	-	rtb-0f28f533b85466aad	-	-	Yes	vpc-0684c34a048585882 Hus...	533267121457
☐	Private-RT	rtb-0858c81532b672ae5	subnet-052475d536a2cc...	-	No	vpc-0684c34a048585882 Hus...	533267121457
☐	Public-RT	rtb-04bdc0fa7a442ac9d	subnet-0514d1738c3c3f...	-	No	vpc-0684c34a048585882 Hus...	533267121457

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.
[Add new tag](#)
You can add 50 more tags.

[Cancel](#) [Create route table](#)

From the left-side menu, select **Route tables** and click on **Create route table**.

Give it a name, choose your VPC, and click **create route table**.

☒ Public-RT2 [rtb-0c4084bf258394293](#) - - No [vpc-0684c34a048585882 | Hus...](#) 533267121457

rtb-0c4084bf258394293 / Public-RT2

Details **Routes** Subnet associations Edge associations Route propagation Tags

Routes (1) Both [Edit routes](#)

Destination	Target	Status	Propagated
192.168.0.0/16	local	Active	No

Select your route table. From the drop-down menu, select Route, and click **Edit routes**.

Edit routes

Destination	Target	Status	Propagated
10.0.0.0/16	local	Active	No

[Add route](#)

[Cancel](#) [Preview](#) [Save changes](#)

Click on **Add route**.

Edit routes

Destination	Target	Status	Propagated
192.168.0.0/16	local	Active	No
Q 0.0.0.0/0	Internet Gateway		No

[Add route](#)
[Cancel](#)
[Preview](#)
[Save changes](#)

First, choose **0.0.0.0/0**, then for the **target**, choose **internet gateway**, and then select the **internet gateway** that you created before. Finally, click **save changes**.

Public RT2	rtb-0c4084bf258394293	-	-	No	ygc-0684c34a048585882 Hus...	533267121457
rtb-0c4084bf258394293 / Public-RT2						
Details Routes Subnet associations Edge associations Route propagation Tags						
Explicit subnet associations (0)						
Find subnet association						
No subnet associations You do not have any subnet associations.						

After you create the route table, you must add the subnets. First, choose the public route table and from the drop-down menu select **Subnet associations**, then choose **Edit subnet associations**.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/3)				
Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
Private	subnet-052475d536a2cc057	192.168.1.0/24	-	rtb-0858c81532b672ae5 / Private-RT
☒ public2	subnet-09b77418612f86141	192.168.3.0/24	-	Main (rtb-0f28f333b85466ead)
Public	subnet-0514d1738c5c3f1d2	192.168.2.0/24	-	rtb-04bdb0fa2a442ac9d / Public-RT

Selected subnets
 subnet-09b77418612f86141 / public2

[Cancel](#)
[Save associations](#)

Choose the **public2** subnet, then click **save associations**.

Services

[Show more](#)

EC2

Virtual Servers in the Cloud



EC2

[Dashboard](#)
[EC2 Global View](#)
[Events](#)

Instances (1/2) [Info](#)

Find Instance by attribute or tag (case-sensitive) [All states](#)

Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☐	Webserver	i-0853bd0bd43844afa	Terminated	t2.micro	-	View alarms	us-east-1b	-	-	-	-
☒	webserver	i-0a779aed065bcca51	Stopped	t2.micro	-	View alarms	us-east-1b	-	-	-	-

Go to EC2 from the side menu, choose **Launch instance**.

Instances

Instances
[Instance Types](#)
[Launch Templates](#)
[Spot Requests](#)
[Savings Plans](#)

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

e.g. My Web Server

[Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

[Recents](#)
[My AMIs](#)
[Quick Start](#)
☒ Owned by me

☐ Shared with me

[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Webserver2

ami-0dbbc2aeaba67a847

2025-04-23T12:47:55.000Z

Virtualization: hvm

ENA enabled: true

Root device type: ebs

Boot mode: uefi-preferred

Description

webserver2

Architecture

x86_64

AMI ID

ami-0dbbc2aeaba67a847

First, give your EC2 a name. Second, in **Application and OS Images (Amazon Machine Image)**, select **My AMIs**. Third, make sure you select **Owned by me**. Lastly, choose the image you created before.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Select

 [Create new key pair](#)

In the Key pair, select the **earlier key you created**.

▼ Network settings [Info](#)

[Edit](#)

Network | [Info](#)

vpc-0f76a9bd63d923656

Subnet | [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP | [Info](#)

Enable

[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere

0.0.0.0/0

☐ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

 Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

×

In the network settings, click on **Edit**.

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-Of76a9bd63d923656 (default) ▼ 

172.31.0.0/16

Subnet [Info](#)

No preference ▼ 

Create new subnet [↗](#)

Auto-assign public IP [Info](#)

Enable ▼

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

launch-wizard-1

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./()#,@[]+=&;!\$*

First, choose your VPC. Second, choose your second private subnet you created before and **make sure** that the **Auto-assign public IP is Enable**. Lastly, click on **Select existing security group** and select your security group, then click on **Launch instance**.

Create Target Group

EC2

Dashboard

EC2 Global View 

Events

► Instances

► Images

► Elastic Block Store

► Network & Security

▼ Load Balancing

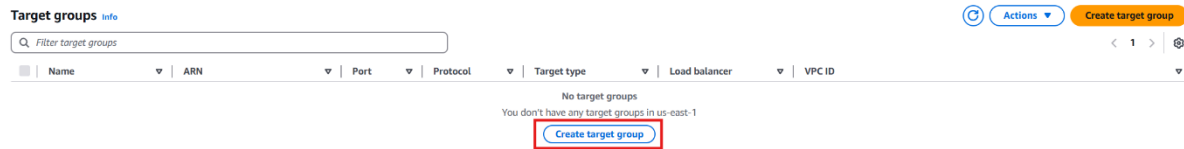
Load Balancers

Target Groups

Trust Stores

▼ Auto Scaling

Auto Scaling Groups



Go to EC2 from the side menu, choose Target group. Click on **Create Target group**.

Specify group details

Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Basic configuration

Settings in this section can't be changed after the target group is created.

Choose a target type

☒ Instances

- Supports load balancing to instances within a specific VPC.
- Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.

☐ IP addresses

- Supports load balancing to VPC and on-premises resources.
- Facilitates routing to multiple IP addresses and network interfaces on the same instance.
- Offers flexibility with microservice based architectures, simplifying inter-application communication.
- Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.

☐ Lambda function

- Facilitates routing to a single Lambda function.
- Accessible to Application Load Balancers only.

☐ Application Load Balancer

- Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
- Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

Target group name

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scroll down and give the target group a name.

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0f76a9bd63d923656
IPv4 VPC CIDR: 172.31.0.0/16

Protocol version

☒ HTTP1
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC
Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Choose your **VPC** that you created before.

Attributes

① Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

► **Tags - optional**
Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

Cancel **Next**

Leave everything as the default option and click Next.

Register targets
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Instance ID	Name	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-0a779aed665bcca51	webserver	Running	Hussain	us-east-1b	192.168.2.39	subnet-0514d1738c5c3f1d2	April 23, 2025, 16:51 (UTC-07:00)
i-0b79139cb2bcc9bfa		Running	Hussain	us-east-1c	192.168.5.187	subnet-09b77418612f86141	April 23, 2025, 16:24 (UTC-07:00)

2 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with comma)

Include as pending below

Review targets

Targets (0)

○ Show only pending

Remove all pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
No instances added yet Specify instances above, or leave the group empty if you prefer to add targets later.								

0 pending

Cancel Previous **Create target group**

Select your two EC2 instances and click on **Include as pending below**. Last, click on **Create Target group**.

Create Load Balancer

EC2

Dashboard

EC2 Global View 

Events

► Instances

► Images

► Elastic Block Store

► Network & Security

▼ Load Balancing

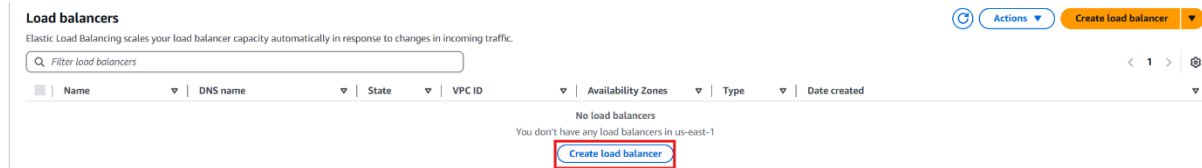
Load Balancers

Target Groups

Trust Stores

▼ Auto Scaling

Auto Scaling Groups



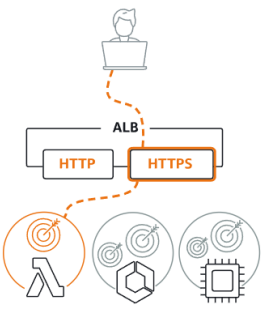
Go to EC2 from the side menu and choose **Load Balancer**. Click on **Create load balancer**.

Compare and select load balancer type

A complete feature-by-feature comparison along with detailed highlights is also available. [Learn more](#)

Load balancer types

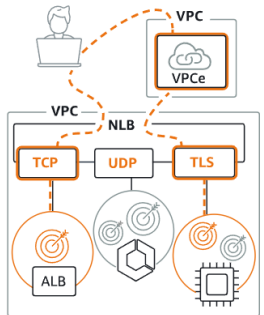
Application Load Balancer [Info](#)



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

Create

Network Load Balancer [Info](#)



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

Create

Gateway Load Balancer [Info](#)



Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

Create

► Classic Load Balancer - *previous generation*

Close

Select **Application Load Balancer** and click on **Create**.

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**
Includes only IPv4 addresses.

☐ **Dualstack**
Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **internet-facing** load balancers only.

Enter a name for the **Load Balancer**.

Network mapping [Info](#)
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#). For a new VPC, [create a VPC](#).

Hussain-VPC
vpc-0684c34a048585882
IPv4 VPC CIDR: 192.168.0.0/16

IP pools - new [Info](#)
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view [Pools in Amazon VPC IP Address Manager console](#).

☐ **Use IPAM pool for public IPv4 addresses**
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☐ **us-east-1a (use1-az2)**

☒ **us-east-1b (use1-az4)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0514d1738c5c3f1d2
IPv4 subnet CIDR: 192.168.2.0/24

us-east-1c (use1-az6)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-09b77418612f86141
IPv4 subnet CIDR: 192.168.3.0/24

First, choose **your VPC** that you created before. Second, under **availability zones and subnets**, select the zones that are associated with public subnets.

Note: For the load balancer to work, it needs two subnets in different availability zones.

Security groups [Info](#)
A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups
Select up to 5 security groups

Hussain
sg-0f9b698f1e4e490eb VPC: vpc-0684c34a048585882

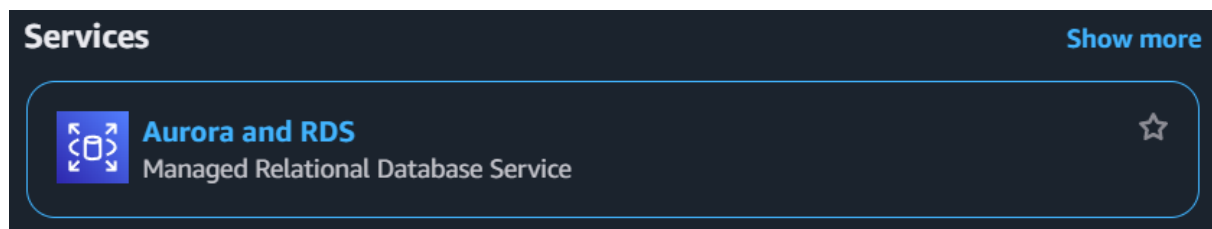
choose your **security group** and **remove the default option**.

How to Create an RDS Database

Note: You must create a [VPC](#) and [EC2 with Ubuntu](#) on it, then you can follow the steps.

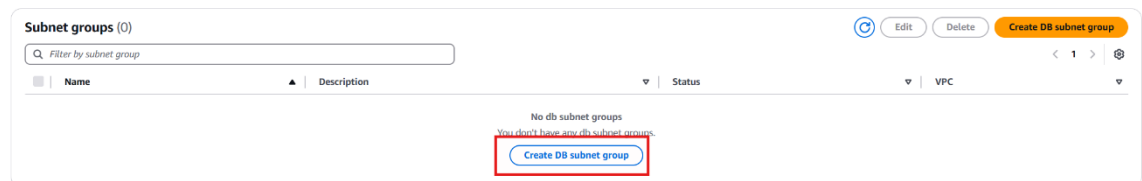
Requirements: that you must have at least **2 Subnets (Public, Private)** and each one must be in a different **Availability Zone**

Create Database Subnet Group



Aurora and RDS

- Dashboard
- Databases
- Query Editor
- Performance insights
- Snapshots
- Exports in Amazon S3
- Automated backups
- Reserved instances
- Proxies
- Subnet groups**
- Parameter groups
- Option groups
- Custom engine versions
- Zero-ETL integrations [New](#)



Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name
You won't be able to modify the name after your subnet group has been created.

Description

VPC
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (0)

Availability zone	Subnet name	Subnet ID	CIDR block
No subnets added to this group			

[Cancel](#)
[Create](#)

First, select the **Subnet group**. Second, click on **Create DB subnet group**. Third, choose a **name** for the subnet group and **write a meaningful Description**. Last, choose your **VPC** that

you created before (and make sure it has at least 2 VPCs and each one in a different Availability Zone).

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone ▲

- ☐ us-east-1a
- ☐ us-east-1b
- ☐ us-east-1c
- ☐ us-east-1d
- ☐ us-east-1e
- ☐ us-east-1f

Subnets selected (0)

Select your **Availability Zones** in which your subnets are in.

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone ▼

us-east-1a ✕ us-east-1b ✕

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets ▲

- ☐ us-east-1a
 - ☐ Public
Subnet ID: subnet-067d80396a9c8d682 CIDR: 192.168.0.0/24
- ☐ us-east-1b
 - ☐ Private
Subnet ID: subnet-00f4b4a8cf2e420a9 CIDR: 192.168.1.0/24

No subnets added to this group

CIDR block

Afterwards, select your Availability Zones that your subnets are in. You will see all the subnets you have in every Availability Zone you choose. Choose the subnets you want them to be in the subnet group.

Note: No need for your subnets to be one public and one private; there can be two privates because you use it for a database, and it does not require public access.

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name
You won't be able to modify the name after your subnet group has been created.

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

VPC
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

2 Subnets, 2 Availability Zones

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Subnet ID: subnet-067d80396a9c8d682 CIDR: 192.168.0.0/24 Subnet ID: subnet-00f4b4a8cf2e420a9 CIDR: 192.168.1.0/24

For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (2)

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1a	Public	subnet-067d80396a9c8d682	192.168.0.0/24
us-east-1b	Private	subnet-00f4b4a8cf2e420a9	192.168.1.0/24

You should have something like this. Finally, click on **create**.

Create Database

Aurora and RDS

Dashboard

Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

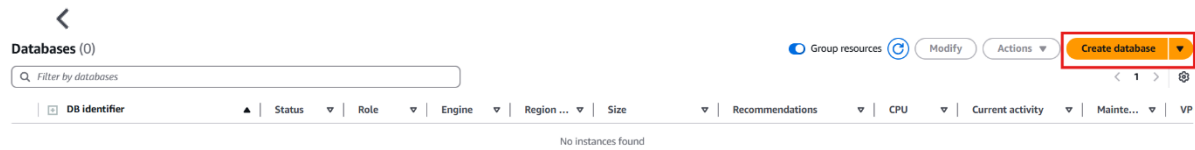
Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations [New](#)



Choose from the side navigator Databases, then choose **Create Database**.

Create database [info](#)

Choose a database creation method

- ☒ **Standard create**
You set all of the configuration options, including ones for availability, security, backups, and maintenance.
- ☐ **Easy create**
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

- Engine type** [info](#)
- ☐ Aurora (MySQL Compatible) 
- ☒ **MySQL** 
- ☐ MariaDB 
- ☐ Microsoft SQL Server 
- ☒ **Aurora (PostgreSQL Compatible)** 
- ☐ PostgreSQL 
- ☐ Oracle 
- ☐ IBM Db2 

First, choose the **standard create** for the database creation method. Second, choose from the **Engine options: MySQL**.

Templates
Choose a sample template to meet your use case.

☐ Production
Use defaults for high availability and fast, consistent performance.

☐ Dev/Test
This instance is intended for development use outside of a production environment.

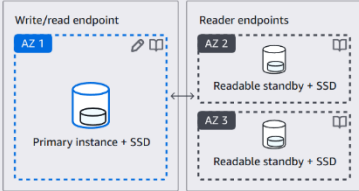
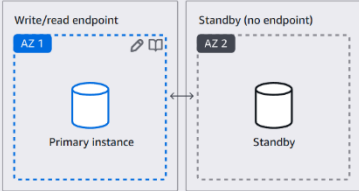
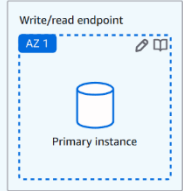
☒ Free tier
Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

Availability and durability
Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. [Learn more in the Amazon RDS service level agreement \(SLA\) \[?\]](#)

☐ Multi-AZ DB cluster deployment (3 instances)
Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:
• 99.95% uptime
• Redundancy across Availability Zones
• Increased read capacity
• Reduced write latency

☐ Multi-AZ DB instance deployment (2 instances)
Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:
• 99.95% uptime
• Redundancy across Availability Zones

☒ Single-AZ DB instance deployment (1 instance)
Creates a single DB instance without standby instances. This setup provides:
• 99.5% uptime
• No data redundancy

Scroll down to **Templates** and choose **Free tier**, and it will automatically choose for you “**Single-AZ DB instance deployment (1 instance)**” in the **Availability and durability**.

Settings

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.
database-1
The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.
admin
1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)
[Password field]
Password strength
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Confirm master password [Info](#)
[Password field]

Scroll down to the settings and change the name for the **DB instance identifier**. Keep the default options and go to the **Master Password**. Enter a strong password and **remember it**.

Note: the password must be at least 8 characters.

Connectivity [Info](#)

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

Network type [Info](#)
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ **IPv4**
Your resources can communicate only over the IPv4 addressing protocol.

☐ **Dual-stack mode**
Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-0f76a9bd65d923656)
6 Subnets, 6 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

[?](#) After a database is created, you can't change its VPC.

Scroll down to **Connectivity** and choose your VPC that you created before.

Public access [Info](#)

☐ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ **Choose existing**
Choose existing VPC security groups

☐ **Create new**
Create new VPC security group

Existing VPC security groups
Choose one or more options

Availability Zone [Info](#)
No preference

For the **VPC security group (firewall)**, choose existing. Remove the default **VPC security groups** and add your **VPC security groups** that you have created before. For the **Availability Zone**, choose it based on your private subnet's **Availability Zone**.

Note: You should **Never select Yes** for **Public Access** only if the question specifies it.

Estimated monthly costs

The Amazon RDS Free Tier is available to you for 12 months. Each calendar month, the free tier will allow you to use the Amazon RDS resources listed below for free:

- 750 hrs of Amazon RDS in a Single-AZ db.t2.micro, db.t3.micro or db.t4g.micro Instance.
- 20 GB of General Purpose Storage (SSD).
- 20 GB for automated backup storage and any user-initiated DB Snapshots.

[Learn more about AWS Free Tier.](#)

When your free usage expires or if your application use exceeds the free usage tiers, you simply pay standard, pay-as-you-go service rates as described in the [Amazon RDS Pricing page](#).

[?](#) You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

[Cancel](#) [Create database](#)

Click on **Create database**.

Note: it will take several minutes to fully create the database, wait for it for the status to change to **Available**.

Connect to your Database

Instances (1/1) Info									
Find Instance by attribute or tag (case-sensitive) All states ▾									
☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...
☒	webServer1	i-0c96dd20eca6f7f51	Running	t2.micro	Initializing	View alarms +	us-east-1b	-	52.91.158.104

Now go to the EC2 instance that you have created before:

To connect to the web server, **choose your instance** and click on **Connect**.

Connect to instance [Info](#)

Connect to your instance i-0c96dd20eca6f7f51 (webServer1) using any of these options

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

Instance ID

i-0c96dd20eca6f7f51 (webServer1)

Connection Type

☒ Connect using EC2 Instance Connect
 Connect using the EC2 Instance Connect browser-based client, with a public IPv4 or IPv6 address.

☐ Connect using EC2 Instance Connect Endpoint
 Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Public IPv4 address

52.91.158.104

IPv6 address

-

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.

Note:

In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

Connect

Choose **EC2 Instance Connect** and click on **Connect**.

```

Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-1024-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Apr 21 15:50:11 UTC 2025

System load: 0.96               Processes: 106
Usage of /: 25.0% of 6.71GiB    Users logged in: 0
Memory usage: 20%              IPv4 address for enx0: 192.168.0.19
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo apt update

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-192-168-0-19:~$ sudo apt update

```

After you connect to the EC2 instance, issue the command “**Sudo apt update**”. After finishing the update, run the following command “**sudo apt install mysql-client**” to install the **mysql-client** in your EC2.

50

```
ubuntu@ip-192-168-0-19:~$ sudo apt install mysql-client
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  mysql-client-8.0 mysql-client-core-8.0 mysql-common
The following NEW packages will be installed:
  mysql-client mysql-client-8.0 mysql-client-core-8.0 mysql-common
0 upgraded, 4 newly installed, 0 to remove and 63 not upgraded.
Need to get 2765 kB of archives.
After this operation, 61.8 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
```

Enter “Y” to continue the installation.

Databases (1)

Filter by databases

DB identifier	Status	Role	Engine	Region ...	Size	Recommendations	CPU	Current activity	Mainte...	VP
database-1	Available	Instance	MySQL Co...	us-east-1b	db.t4g.micro		3.21%	0 Connections	none	vp

To connect to the database that you have created before, go to **Aurora and RDS** and click on the name of your database.

database-1

Summary

DB identifier database-1	Status Available	Role Instance	Engine MySQL Community	Recommendations
CPU 3.57%	Class db.t4g.micro	Current activity 0 Connections	Region & AZ us-east-1b	

Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integrations | Maintenance & backups | Data migrations - new | Tags | Recommendations

Connectivity & security

Endpoint & port Endpoint database-1.ch40cawco4od.us-east-1.rds.amazonaws.com Port 3306	Networking Availability Zone us-east-1b VPC Hussain-VPC (vpc-055203bfff6b96b89) Subnet group polytechnic-rds Subnets subnet-067d80396a9c8d682 subnet-00f4b4a8cf2e420a9 Network type IPv4	Security VPC security groups Hussain-SEC (sg-04b269ca39616c755) Active Publicly accessible No Certificate authority rds-ca-rsa2048-g1 Certificate authority date May 26, 2061, 02:34 (UTC+03:00) DB Instance certificate expiration date April 21, 2026, 19:27 (UTC+03:00)
---	--	--

Copy The **Endpoint** and remember the **port number**.

```
ubuntu@ip-192-168-0-19:~$ sudo su
root@ip-192-168-0-19:/home/ubuntu#
```

Enter in the terminal “**sudo su**” to go to the root.

```
root@ip-192-168-0-19:/home/ubuntu# mysql -h database-1.ch40cawco4od.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 31
Server version: 8.0.40 Source distribution

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

Now you are at the root issue. The command “**mysql -h <Endpoint> -P <port number> -u admin -p**” will connect you to the MySQL database that you created. It will ask you to enter the password you chose when creating the database.

If it is shown **mysql>** in the end that means that you are in the database, you did correct.

Note: the endpoint and the port number are in the previous screenshot.

Now you can test how SQL work, try to enter this command for a test:

```
create database employees;
```

```
use employees;
```

```
create table poly(studentid int, studname varchar(20), marks int);
```

```
insert into poly values(1,'Ali', 30);
```

```
select * from poly;
```

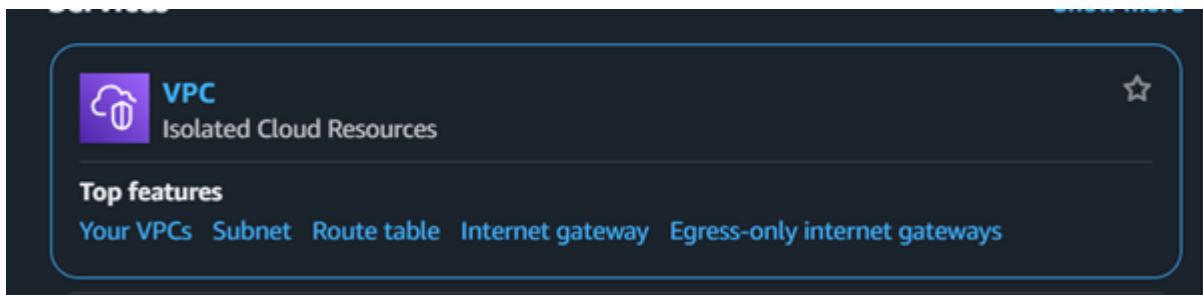

How to Create NAT Gateways

Note: you must create a [VPC](#) and [two EC2 instances with Ubuntu](#) on them (one with the public subnet and the other with the private subnet) then you can follow the steps.

Remember:

- ❖ You must have at least **2 Subnets (Public, Private)**, and each one must be in a different **Availability Zone**.
- ❖ The EC2 that you will create using the private subnet, you must **disable the Auto-assign public IP**.

Create NAT Gateways



VPC dashboard

EC2 Global View

Filter by VPC:

- Virtual private cloud
- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only Internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways**
- Peering connections
- Route servers

NAT gateways

Find NAT gateways by attribute or tag

Name	NAT gateway ID	Connectivity...	State	State message	Primary public I...	Primary private I...	Primary network...	VPC
No NAT gateways found								

Create NAT gateway

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Subnet
Select a subnet in which to create the NAT gateway.

Connectivity type
Select a connectivity type for the NAT gateway.
☒ Public
☐ Private

Elastic IP allocation ID
Assign an Elastic IP address to the NAT gateway.

Additional settings

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.

You can add 50 more tags.

Cancel

First, go to **VPC** from the left-side menu, choose **NAT Gateways**, and click **Create NAT Gateway**. Second, enter a name for the NAT gateway and choose the **public subnet**. Third, click on **Allocate Elastic IP**. Last, click on **Create NAT gateway**.

Note: the reason we create a NAT gateway is that the NAT gateway is a one-way direction that allows the service to access the internet, but doesn't allow the internet to access the service. It is a very good choice if you want to deploy an update or a batch for a database, for example.

Link the NAT Gateway with a Private Subnet

VPC dashboard <

[EC2 Global View](#)

Filter by VPC: ▼

▼ **Virtual private cloud**

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only Internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers [New](#)

Route tables (1/4) info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
-	rtb-0208bdfb0c0ff0a0b8	-	-	Yes	vpc-0f76a9bd63d923656	533267121457
-	rtb-0f0920d21d757aec8	-	-	Yes	vpc-0cc38ae375bd66789 Huss...	533267121457
Public-RT	rtb-075c6cc602e397393	subnet-053b19ebb70cac...	-	No	vpc-0cc38ae375bd66789 Huss...	533267121457
Private-RT	rtb-0454740d316764879	subnet-067b18d4adecd...	-	No	vpc-0cc38ae375bd66789 Huss...	533267121457

rtb-0454740d316764879 / Private-RT

Details **Routes** Subnet associations Edge associations Route propagation Tags

Routes (1)

Destination	Target	Status	Propagated
192.168.0.0/16	local	Active	No

Go to **Routing tables** and choose your **private routing table**, and from the drop-down menu, choose **Routes** and click on **Edit routes**.

Edit routes

Destination	Target	Status	Propagated
192.168.0.0/16	local	Active	No

[Add route](#)

Cancel [Preview](#) [Save changes](#)

Edit routes

Destination
192.168.0.0/16

Add route

Target
local

Status
Active

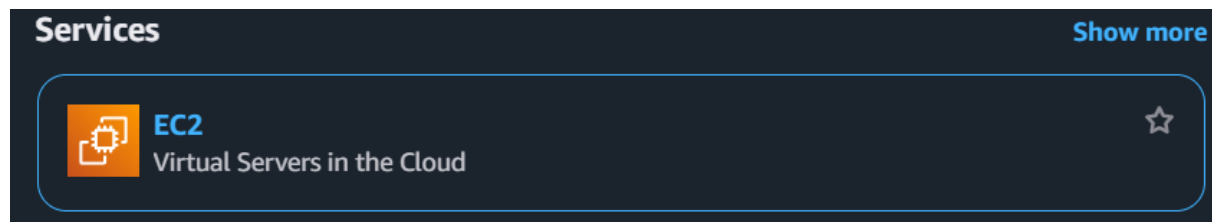
Propagated
No

Remove

Cancel Preview **Save changes**

In Edit routes, click on **Add route** and choose **0.0.0.0/0** for the **destination**, and for the **Target**, choose **NAT Gateway** and **select your NAT gateway** you created before, and click on **Save changes**.

Checking the NAT gateway is working in the Private Server



EC2

- Dashboard
- EC2 Global View
- Events
- Instances
 - Instances**
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
- Images
 - AMIs
 - AMI Catalog

Instances (2) info

All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
private server	i-069221c1cc36011a9	Running	t2.micro	Initializing	View alarms +	us-east-1a	-	-	-	-
public server	i-0096093d5f5a0268	Running	t2.micro	Initializing	View alarms +	us-east-1b	-	54.146.150.123	-	-

Go to EC2 and from the left-side menu choose **instances**. You should have **two instances running**, one with a public subnet and the other with a private subnet.

Note: You can use different keys and security groups for each instance, but note that **you must remember** what key you use for each instance.

Instances (1/2) [Info](#)

Last updated 3 minutes ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) [All states](#)

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☐	private server	i-069221c1cc36011a9	Running	t2.micro	Initializing	View alarms +	us-east-1a	-	-	-	-
☒	public server	i-0096093d5f6ac0268	Running	t2.micro	Initializing	View alarms +	us-east-1b	-	54.146.150.123	-	-

Connect [Info](#)

Connect to an instance using the browser-based client.

EC2 Instance Connect Session Manager **SSH client** EC2 serial console

Instance ID
i-0096093d5f6ac0268 (public server)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is `Hussain-key.pem`.
3. Run this command, if necessary, to ensure your key is not publicly viewable.
`chmod 400 "Hussain-key.pem"`
4. Connect to your instance using its Public IP:
54.146.150.123

Example:
`ssh -i "Hussain-key.pem" ubuntu@54.146.150.123`

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

[Cancel](#)

Now select your instance that has the **public subnet** on it and click on **Connect**. Click on **SSH client** and copy the Example.

```

Microsoft Windows [Version 10.0.26100.3194]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hussa>
  
```

Open the **CMD** and go to where you saved your Key, and **paste the command you copied** from the AWS.

```
Command Prompt - ssh -i "H x + v
Microsoft Windows [Version 10.0.26100.3194]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hussa>cd Downloads

C:\Users\hussa\Downloads>ssh -i "Hussain-key.pem" ubuntu@54.146.150.123
The authenticity of host '54.146.150.123 (54.146.150.123)' can't be established.
ED25519 key fingerprint is SHA256:PzJFeHnXHfnbhsBDNW5Vw7s6pq1WfCWjJ8k2J1jI+Pw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes|
```

It will ask you to continue the connection by entering “YES”.

```
ubuntu@ip-192-168-1-226: ~ x + v
System information as of Sun Apr 27 15:02:01 UTC 2025

System load:  0.0      Processes:      104
Usage of /:   25.0% of 6.71GB  Users logged in:  0
Memory usage: 20%      IPv4 address for enX0: 192.168.1.226
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

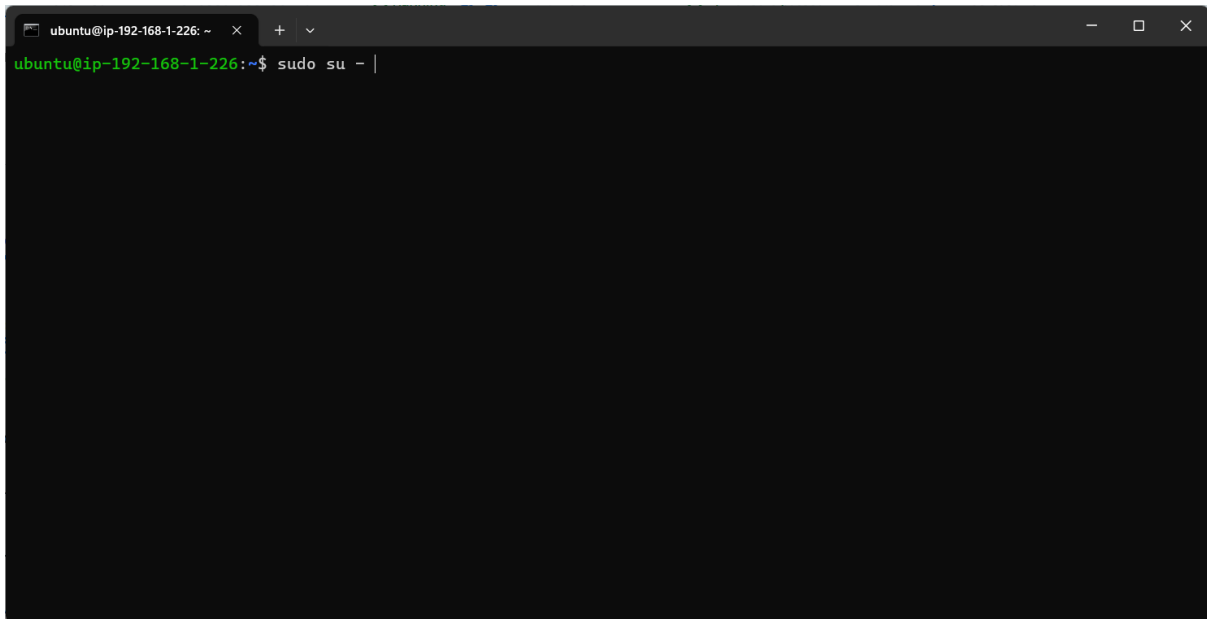
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-192-168-1-226:~$ |
```

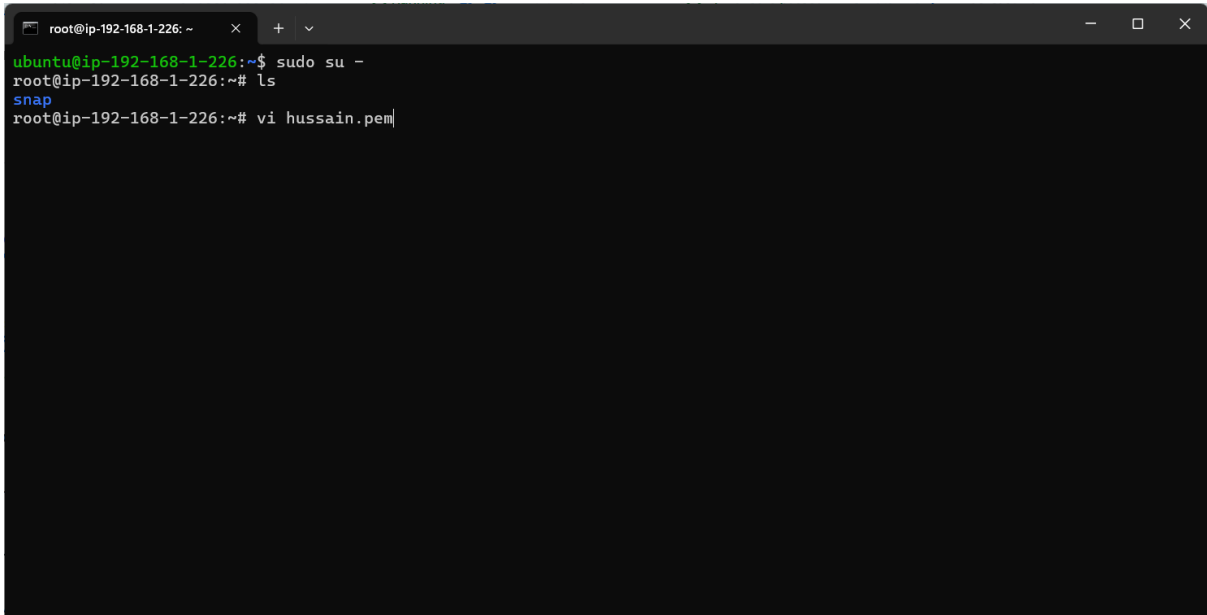
If you have done the EC2 instance correctly, it will open for you like this.

Now update your Ubuntu by typing the command “**sudo apt update**” and install Apache2 by running the command “**sudo apt install apache2**”.

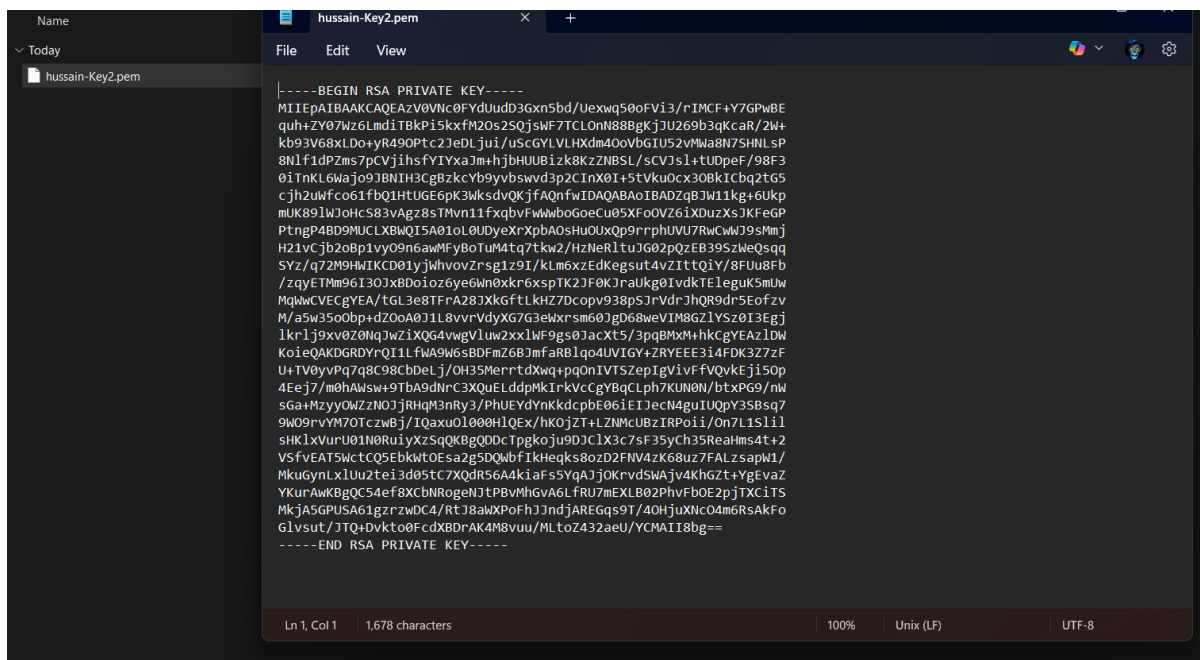
Connect to the Private Server using the Public Server.

A terminal window with a dark background. The title bar shows "ubuntu@ip-192-168-1-226: ~". The prompt is "ubuntu@ip-192-168-1-226:~\$". The command "sudo su -" has been entered, and a cursor is visible at the end of the line.

Now go to the Root by running the command “**sudo su -**”

A terminal window with a dark background. The title bar shows "root@ip-192-168-1-226: ~". The prompt is "root@ip-192-168-1-226:~#". The command "sudo su -" has been entered, followed by "ls" and "vi hussain.pem". The prompt has changed to "root@ip-192-168-1-226:~#".

In the root, issue the command “**Vi and any name. pem**” to create a file in which you will put the key for a private EC2 instance.

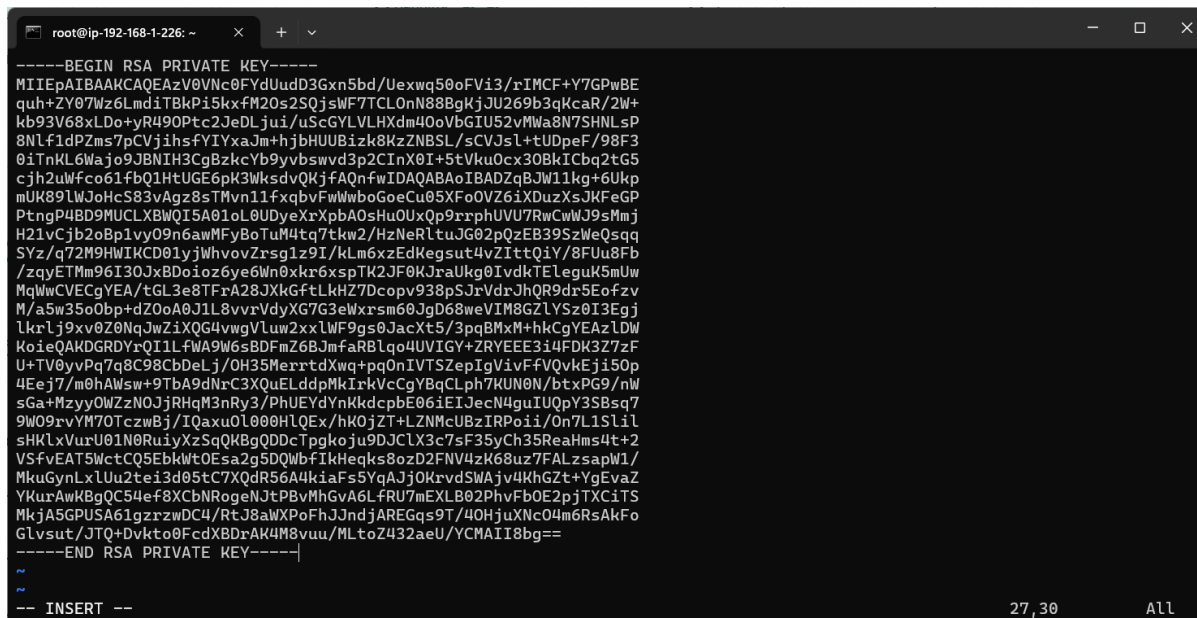


```

-----BEGIN RSA PRIVATE KEY-----
MIIEpAIBAAKCAQEAzV0Vnc0FYdUudD3Gxn5bd/Uexwq50oFVi3/rIMCF+Y7GPwBE
quh+ZY07Wz6LmdiTBkPi5kxfM20s2S0jsWf7TCL0nN88BgKjJU269b3qKcaR/2W+
kb93V68xLDo+yR490Ptc2JeDLjui/uScGYLVHXdM40oVbGIU52vMwa8N7SHNLsP
8Nlf1dPZms7pCVjihsfYIYxaJm+hjBUUBizk8KzZNBLS/sCVJsl+tUDpeF/98F3
0iTNkL6Wajo9JBNIH3CgBzkcYb9yvbSwvd3p2CInX0I+5tVku0cx30BkICbq2tG5
cjh2uWfco61fbQ1HTUG6pK3WksdvQKjFAQnfWIDAQABaoIBADZqBJW11kg+6Ukp
mUK89LWJoHcS83vAgz8sTMvn11fxqbfvFwWwboGoeCu05Xfo0VZ6iXDuzXsJKfEGP
PtngP4BD9MUCLXBWQI5A01oL0UDyeXrXpbA0sHuOUxQp9rrrphUVU7RwCwWJ9sMmj
H21vCjb2oBp1vy09n6awMFyBoTuM4tq7tkw2/HzNeRltuJG02pQzEB39S2WeQsqQ
SYz/q72M9HWIKCD01yJwhvovZrsg1z9I/kLm6xzEdKegsut4vZiTTQ1Y/8FUu8Fb
/zqyETMm96I30JxBDoioz6ye6Wn0xkr6xspTK2JF0KJraUkg0IvdKTEleguK5mUw
MqWwCVCeGYEA/tGL3e8TFrA28JXkGfTLkHZ7Dcopv938pSjrVdrJhQR9dr5Eofzv
M/a5w35o0bp+dZ0oA0J1L8vvrVdyXG7G3eWxrm60JgD68weVIM8GZLYS20I3EgJ
lkr1j9xv0Z0NqJwZiXQG4vWgVluw2xxLWF9gs0JacXt5/3pqBMxM+hkCGYEAZLDW
KoieQAKDGRDYrQI1LFWA9W6sBDFmZ6BjmfARBlqo4UVIGY+ZRYEE3i4FDK3Z7zf
U+TV0yvPq7q8C98CbDeLj/OH35MerrtdXwq+pq0nIVTSZepIgvVivfVQvKEji50p
4Eej7/m0hAWsw+9TbA9dNrC3XQuELddpMkIrKvCcGYBqCLph7KUN0N/btxPG9/nW
sGa+Mzyy0WZzNOjJRhgM3nRy3/PhUEYdYnKkdcPbE06iEIJecN4guIUQpY3SBsq7
9W09rvYM70TczwBj/IQaxu0L000HLQEx/hK0jZT+LZNMcuBzIRPoii/On7L1Sl1l
sHK1xVurU01N0RuiYXzSgQK8GQDDcTpgkoju9DJCLX3c7sF35yCh35ReaHms4t+2
VSfVEAT5wctCQ5EbkWt0Esa2g5DQWbfiKHeqks8ozD2FNv4zK68uz7FALZsapW1/
MkuGynLxLUu2tei3d05tC7XQdr56A4kiaFs5YqAJjOKrVdSWAjbv4KhGZt+YgEvaZ
YkurAwkBgQC54ef8XCbNRogeNjtpBvMhGvA6LFRU7mEXLB02PhvFb0E2pJTXCiTS
MkJA5GPUSA61gzrzwDC4/RtJ8aWXPoFhJJndJAREGqs9T/40HjuXNc04m6RsAKFo
GLvsut/JTQ+Dvkt0FcdXBDrAK4M8vuu/MLtoZ432aeU/YCMAII8bg==
-----END RSA PRIVATE KEY-----

```

Go to your key, which is for the private instance, and open it with a notepad and copy the whole text.



```

-----BEGIN RSA PRIVATE KEY-----
MIIEpAIBAAKCAQEAzV0Vnc0FYdUudD3Gxn5bd/Uexwq50oFVi3/rIMCF+Y7GPwBE
quh+ZY07Wz6LmdiTBkPi5kxfM20s2S0jsWf7TCL0nN88BgKjJU269b3qKcaR/2W+
kb93V68xLDo+yR490Ptc2JeDLjui/uScGYLVHXdM40oVbGIU52vMwa8N7SHNLsP
8Nlf1dPZms7pCVjihsfYIYxaJm+hjBUUBizk8KzZNBLS/sCVJsl+tUDpeF/98F3
0iTNkL6Wajo9JBNIH3CgBzkcYb9yvbSwvd3p2CInX0I+5tVku0cx30BkICbq2tG5
cjh2uWfco61fbQ1HTUG6pK3WksdvQKjFAQnfWIDAQABaoIBADZqBJW11kg+6Ukp
mUK89LWJoHcS83vAgz8sTMvn11fxqbfvFwWwboGoeCu05Xfo0VZ6iXDuzXsJKfEGP
PtngP4BD9MUCLXBWQI5A01oL0UDyeXrXpbA0sHuOUxQp9rrrphUVU7RwCwWJ9sMmj
H21vCjb2oBp1vy09n6awMFyBoTuM4tq7tkw2/HzNeRltuJG02pQzEB39S2WeQsqQ
SYz/q72M9HWIKCD01yJwhvovZrsg1z9I/kLm6xzEdKegsut4vZiTTQ1Y/8FUu8Fb
/zqyETMm96I30JxBDoioz6ye6Wn0xkr6xspTK2JF0KJraUkg0IvdKTEleguK5mUw
MqWwCVCeGYEA/tGL3e8TFrA28JXkGfTLkHZ7Dcopv938pSjrVdrJhQR9dr5Eofzv
M/a5w35o0bp+dZ0oA0J1L8vvrVdyXG7G3eWxrm60JgD68weVIM8GZLYS20I3EgJ
lkr1j9xv0Z0NqJwZiXQG4vWgVluw2xxLWF9gs0JacXt5/3pqBMxM+hkCGYEAZLDW
KoieQAKDGRDYrQI1LFWA9W6sBDFmZ6BjmfARBlqo4UVIGY+ZRYEE3i4FDK3Z7zf
U+TV0yvPq7q8C98CbDeLj/OH35MerrtdXwq+pq0nIVTSZepIgvVivfVQvKEji50p
4Eej7/m0hAWsw+9TbA9dNrC3XQuELddpMkIrKvCcGYBqCLph7KUN0N/btxPG9/nW
sGa+Mzyy0WZzNOjJRhgM3nRy3/PhUEYdYnKkdcPbE06iEIJecN4guIUQpY3SBsq7
9W09rvYM70TczwBj/IQaxu0L000HLQEx/hK0jZT+LZNMcuBzIRPoii/On7L1Sl1l
sHK1xVurU01N0RuiYXzSgQK8GQDDcTpgkoju9DJCLX3c7sF35yCh35ReaHms4t+2
VSfVEAT5wctCQ5EbkWt0Esa2g5DQWbfiKHeqks8ozD2FNv4zK68uz7FALZsapW1/
MkuGynLxLUu2tei3d05tC7XQdr56A4kiaFs5YqAJjOKrVdSWAjbv4KhGZt+YgEvaZ
YkurAwkBgQC54ef8XCbNRogeNjtpBvMhGvA6LFRU7mEXLB02PhvFb0E2pJTXCiTS
MkJA5GPUSA61gzrzwDC4/RtJ8aWXPoFhJJndJAREGqs9T/40HjuXNc04m6RsAKFo
GLvsut/JTQ+Dvkt0FcdXBDrAK4M8vuu/MLtoZ432aeU/YCMAII8bg==
-----END RSA PRIVATE KEY-----

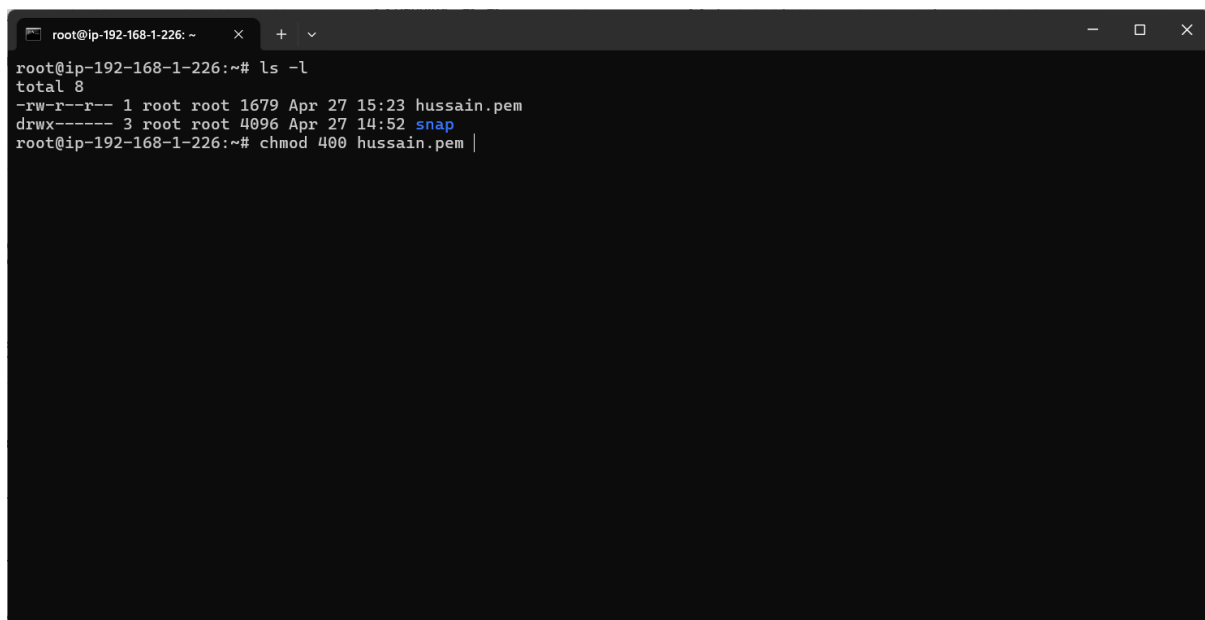
```

Inside your file, you create, enter “i” to insert in the file, paste the key content in it, and click “ESC: wq” to close it and save what you entered.

```
-rw-r--r-- 1 root root 1679 Apr 27 15:23 hussain.pem  
drwx----- 3 root root 4096 Apr 27 14:52 snap  
root@ip-192-168-1-226:~# |
```

If you run the command “**ls -l**”, you will see the key has read permissions for the group and others, and we do not want that.

Note: If we kept permissions like this, anyone can read and write in the key file, and we don’t want that to happen.



```
root@ip-192-168-1-226:~# ls -l  
total 8  
-rw-r--r-- 1 root root 1679 Apr 27 15:23 hussain.pem  
drwx----- 3 root root 4096 Apr 27 14:52 snap  
root@ip-192-168-1-226:~# chmod 400 hussain.pem |
```

By running the command “**chmod 400 <name of the key>.pem**” it will change the permissions.

Note: After running the command, it will change the permissions of the key so that only the owner can read and modify the file.


```

root@ip-192-168-1-226: ~
root@ip-192-168-1-226:~# ls -l
total 8
-rw-r--r-- 1 root root 1679 Apr 27 15:23 hussain.pem
drwx----- 3 root root 4096 Apr 27 14:52 snap
root@ip-192-168-1-226:~# chmod 400 hussain.pem
root@ip-192-168-1-226:~# ls
hussain.pem snap
root@ip-192-168-1-226:~# ls -l
total 8
-r----- 1 root root 1679 Apr 27 15:23 hussain.pem
drwx----- 3 root root 4096 Apr 27 14:52 snap
root@ip-192-168-1-226:~#

```

As you can see, the **permission has been changed**.

Instances (1/2) Info

Find Instance by attribute or tag (case-sensitive) Running

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☒ Private-web	i-01b8eb1a5319014d1	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	-	-	-
☐ public server	i-0096093d5f6ac0268	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	-	54.146.150.123	-	-

i-01b8eb1a5319014d1 (Private-web)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary info

Instance ID i-01b8eb1a5319014d1	Public IPv4 address -	Private IPv4 addresses 192.168.0.190
IPv6 address -	Instance state Running	Public IPv4 DNS -

Go to your **private EC2 instance**, select it, and **copy** from the drop-down menu the **private IP**.

```
ubuntu@ip-192-168-0-190: ~  
System information as of Sun Apr 27 15:31:18 UTC 2025  
  
System load:  0.0      Processes:      103  
Usage of /:   25.0% of 6.71GB  Users logged in:  0  
Memory usage: 20%      IPv4 address for enX0: 192.168.0.190  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
ubuntu@ip-192-168-0-190:~$ |
```

Run the command “**ssh -i <name of the key>.pem ubuntu@<private ip that you copy>**”

```
ED25519 key fingerprint is SHA256:PzJFeHnXHfmbhsBDNWSVw7s6pq1WfCWjJ8k2J1ji+Pw.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes|
```

It will ask you to continue connecting, type: “**YES**”

Note: If you copy the key correctly and **change the permission**, and with the correct **private IP**, it will open like the picture.

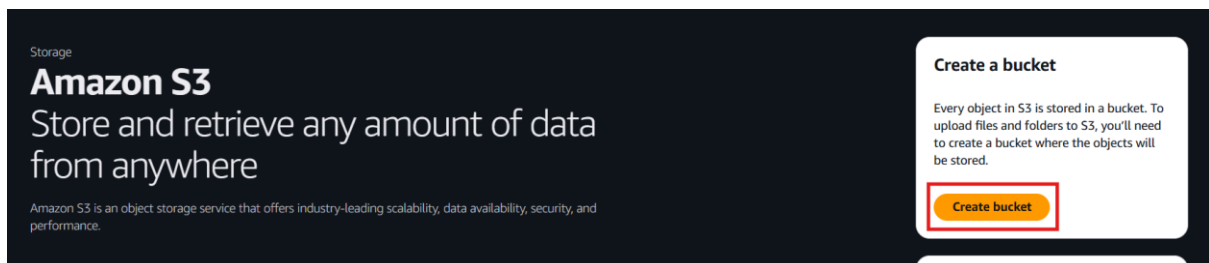
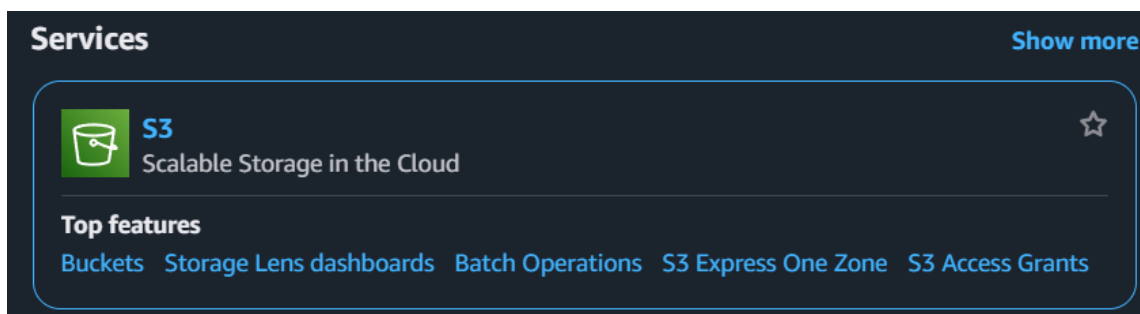
```
ubuntu@ip-192-168-0-190: ~  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
ubuntu@ip-192-168-0-190:~$ ping google.com  
PING google.com (64.233.180.113) 56(84) bytes of data:  
64 bytes from on-in-f113.1e100.net (64.233.180.113): icmp_seq=1 ttl=53 time=3.30 ms  
64 bytes from on-in-f113.1e100.net (64.233.180.113): icmp_seq=2 ttl=53 time=3.10 ms  
64 bytes from on-in-f113.1e100.net (64.233.180.113): icmp_seq=3 ttl=53 time=3.00 ms  
64 bytes from on-in-f113.1e100.net (64.233.180.113): icmp_seq=4 ttl=53 time=2.76 ms  
64 bytes from on-in-f113.1e100.net (64.233.180.113): icmp_seq=5 ttl=53 time=2.69 ms  
64 bytes from on-in-f113.1e100.net (64.233.180.113): icmp_seq=6 ttl=53 time=3.15 ms  
64 bytes from on-in-f113.1e100.net (64.233.180.113): icmp_seq=7 ttl=53 time=2.96 ms  
64 bytes from on-in-f113.1e100.net (64.233.180.113): icmp_seq=8 ttl=53 time=2.88 ms  
64 bytes from on-in-f113.1e100.net (64.233.180.113): icmp_seq=9 ttl=53 time=3.47 ms
```

Run the command “**ping google.com**”. If the ping is successful, that means you have done everything correctly and the **private EC2 instance has internet connection**, but no one can have access from the internet to your private server because of the NAT gateway.

How to Create an S3 Bucket

The S3 bucket is for storing any type of data, and what you store on it is called objects, and it can hold nearly unlimited data. There are many types of it, each one has its benefits, but for this demonstration, we will use the standard one.

Create an S3 bucket without ACL.



First, search for **S3** in the AWS console, in the home page on the S3 bucket, click on **Create bucket**.

Create bucket [Info](#)
Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)
myawsbucket

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn More](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Leave the **bucket type** as the default option, then choose a name for your bucket. It must be unique.

Object Ownership Info
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced

For the **object ownership**, leave the **ACL disabled**.

Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☒ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☒ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☒ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☒ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

For the **Block Public Access settings for this bucket**, uncheck the **Block all public access "** option.

Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Turning off block all public access might result in this bucket and the objects within becoming public
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☒ **I acknowledge that the current settings might result in this bucket and the objects within becoming public.**

Select the **I acknowledge**

Note: by default, the S3 bucket is blocking public access so that you don't have your object in the public, but if you want to, you have to enable the public access.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

- ☒ Disable
☐ Enable

The meaning of **Versioning** is that every update you do for the bucket, AWS will keep a copy of your old version of the bucket, but for this demonstration, you will not use it.

Default encryption

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type

- ☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the [Storage tab of the Amazon S3 pricing page](#).

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

- ☐ Disable
☒ Enable

Advanced settings

After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

[Cancel](#)

[Create bucket](#)

Click on **Create bucket**.

Successfully created bucket "solohunter21"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

Account snapshot - updated every 24 hours All AWS Regions [View Storage Lens dashboard](#)
Storage Lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets | Directory buckets

General purpose buckets (1) All AWS Regions [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Name	AWS Region	IAM Access Analyzer	Creation date
solohunter21	US East (N. Virginia) us-east-1	View analyzer for us-east-1	May 5, 2025, 00:36:24 (UTC+03:00)

click on your bucket name.

solohunter21 Info

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [Access Points](#)

Objects (0) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Name	Type	Last modified	Size	Storage class
No objects You don't have any objects in this bucket.				

[Upload](#)

click on **upload** to upload your files or folders.

Upload [info](#)

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (0)
Remove
Add files
Add folder

All files and folders in this table will be uploaded.

< 1 >

	Name	Folder	Type	Size
No files or folders You have not chosen any files or folders to upload.				

Destination [info](#)

Destination
[s3://solohunter21](#)

► **Destination details**
 Bucket settings that impact new objects stored in the specified destination.

► **Permissions**
 Grant public access and access to other AWS accounts.

► **Properties**
 Specify storage class, encryption settings, tags, and more.

Cancel
Upload

choose **Add files** or **folder** as you want.

Upload [info](#)

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (1 total, 26.5 KB)
Remove
Add files
Add folder

All files and folders in this table will be uploaded.

< 1 >

	Name	Folder	Type	Size
☐	amazon_web_services_73a55e2b43.png	-	image/png	26.5 KB

Destination [info](#)

Destination
[s3://solohunter21](#)

► **Destination details**
 Bucket settings that impact new objects stored in the specified destination.

► **Permissions**
 Grant public access and access to other AWS accounts.

► **Properties**
 Specify storage class, encryption settings, tags, and more.

Cancel
Upload

After uploading your file and folders click on **upload**.

solohunter21 [info](#)

[Objects](#)
[Metadata](#)
[Properties](#)
[Permissions](#)
[Metrics](#)
[Management](#)
[Access Points](#)

Objects (1/1)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

< 1 >

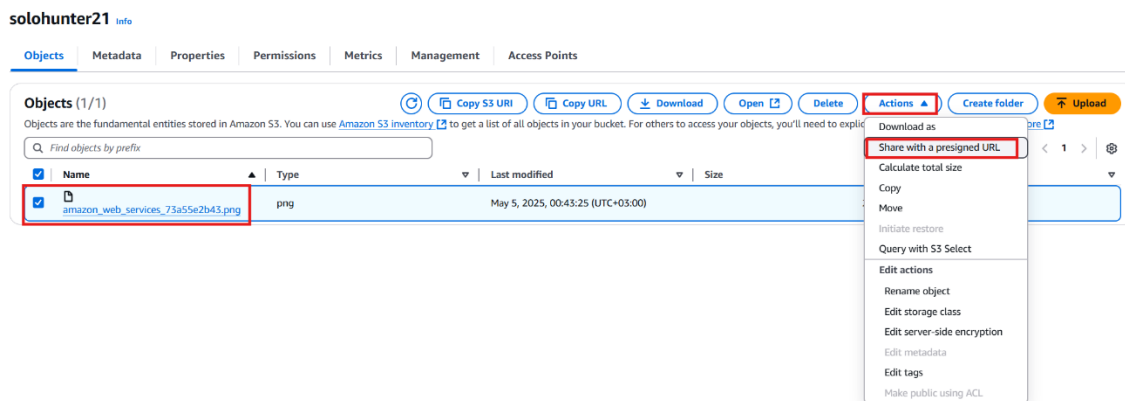
	Name	Type	Last modified	Size	Storage class
☒	amazon_web_services_73a55e2b43.png	png	May 5, 2025, 00:43:25 (UTC+03:00)	26.5 KB	Standard

Return to your bucket and **select your file** that you upload and click on copy URL and open new tab and paste the URL and **see what happen**.

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<Error>
  <Code>AccessDenied</Code>
  <Message>Access Denied</Message>
  <RequestId>XX7MCAPGKNJWE920</RequestId>
  <HostId>xV2+6D93DQynY8k2yy2gMz/n8qQvz5g8duJ9Xg4Mxr5HCc+/CuzuGsZq/6kdsKfBGvDTns5j+ic=</HostId>
</Error>
```

It will appear like this. The reason the photo won't open is that you gave public access to the bucket, not the file itself. If you want to access the files inside the bucket, you need to specify the exact file you want to access.



To give access for the photo first, **select the photo**. Second, click on **action**, then on **share with a presigned URL**.

Share "amazon_web_services_73a55e2b43.png" with a presigned URL

Presigned URLs are used to grant access to an object for a limited time. [Learn more](#)

Anyone can access the object with this presigned URL until it expires, even if the bucket, and object are private.

Time interval until the presigned URL expires

Using the S3 console, you can share an object with a presigned URL for up to 12 hours or until your session expires. To create a presigned URL with a longer time interval, use the AWS CLI or AWS SDK. Time intervals for presigned URLs can be restricted by your IAM policy.

☒ Minutes
☐ Hours

Number of minutes

Must be a whole number between 1 and 720.

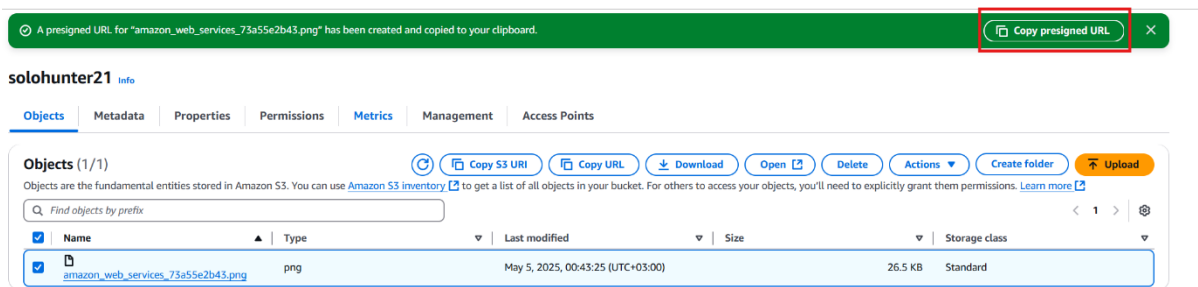
After you create the presigned URL, it's automatically copied to your clipboard.

Cancel

Create presigned URL

Choose how much time you want the photo to be accessible, and click on **Create presigned URL**.

68



Click on **copy presigned URL** and paste it in a new tab.

If you don't have this, you can click on **open**, and it will open the photo in a new tab.



As you can see, the photo can now be accessed.

Create an S3 bucket with ACL

Storage

Amazon S3

Store and retrieve any amount of data from anywhere

Amazon S3 is an object storage service that offers industry-leading scalability, data availability, security, and performance.

Create a bucket

Every object in S3 is stored in a bucket. To upload files and folders to S3, you'll need to create a bucket where the objects will be stored.

Create bucket

First, search for the S3 bucket, then in the home page of the S3 bucket, click on **create bucket**.

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)

myawsbucket

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn More](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Leave the **bucket type as default choice**, then choose a name for your bucket. **It must be unique.**

Object Ownership

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

☐ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☒ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

⚠️ We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users outside of your account simplifies permissions management and auditing.

Object Ownership

☒ **Bucket owner preferred**
If new objects written to this bucket specify the bucket-owner-full-control canned ACL, they are owned by the bucket owner. Otherwise, they are owned by the object writer.

☐ **Object writer**
The object writer remains the object owner.

📌 If you want to enforce object ownership for new objects only, your bucket policy must specify that the bucket-owner-full-control canned ACL is required for object uploads. [Learn more](#)

Now you will choose **ACL enable** and leave the **object Ownership** as the **default choice**.

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ Block all public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☒ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☒ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☒ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☒ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

For the **Block Public Access settings for this bucket**, uncheck the Block all public access.

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ Block all public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

⚠ Turning off block all public access might result in this bucket and the objects within becoming public

AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

Select the **I acknowledge**.

Default encryption [Info](#)

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

- ☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
 - ☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
 - ☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
- Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the [Amazon S3 pricing page](#).

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

- ☐ Disable
- ☒ Enable

Advanced settings

After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

[Cancel](#)

[Create bucket](#)

Leave the rest as the default option and click on Create bucket.

✓ Successfully created bucket "solohunter21"
[View details](#) ✕

To upload files and folders, or to configure additional bucket settings, choose [View details](#).

▶ **Account snapshot** - updated every 24 hours All AWS Regions [View Storage Lens dashboard](#)

solohunter21 [Info](#)

Objects
Metadata
Properties
Permissions
Metrics
Management
Access Points

Objects (0)
[Copy S3 URI](#)
[Copy URL](#)
[Download](#)
[Open](#)
[Delete](#)
[Actions](#)
[Create folder](#)
[Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

☐	Name	▲	Type	▼	Last modified	▼	Size	▼	Storage class	▼
No objects You don't have any objects in this bucket.										

[Upload](#)

click on your bucket name.

Click on **upload** to upload your files or folders.

choose Add files or folder as you want.

Upload
[Info](#)

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (1 total, 26.5 KB)
 [Remove](#)
[Add files](#)
[Add folder](#)

All files and folders in this table will be uploaded.

☐	Name	Folder	Type	Size
☐	amazon_web_services_73a55e2b43.png	-	image/png	26.5 KB

Destination
[Info](#)

Destination
[s3://solohunter21](#)

► Destination details
 Bucket settings that impact new objects stored in the specified destination.

► Permissions
 Grant public access and access to other AWS accounts.

► Properties
 Specify storage class, encryption settings, tags, and more.

[Cancel](#)
[Upload](#)

Destination
[Info](#)

Destination
[s3://solohunter21](#)

► Destination details
 Bucket settings that impact new objects stored in the specified destination.

► Permissions
 Grant public access and access to other AWS accounts.

► Properties
 Specify storage class, encryption settings, tags, and more.

[Cancel](#)
[Upload](#)

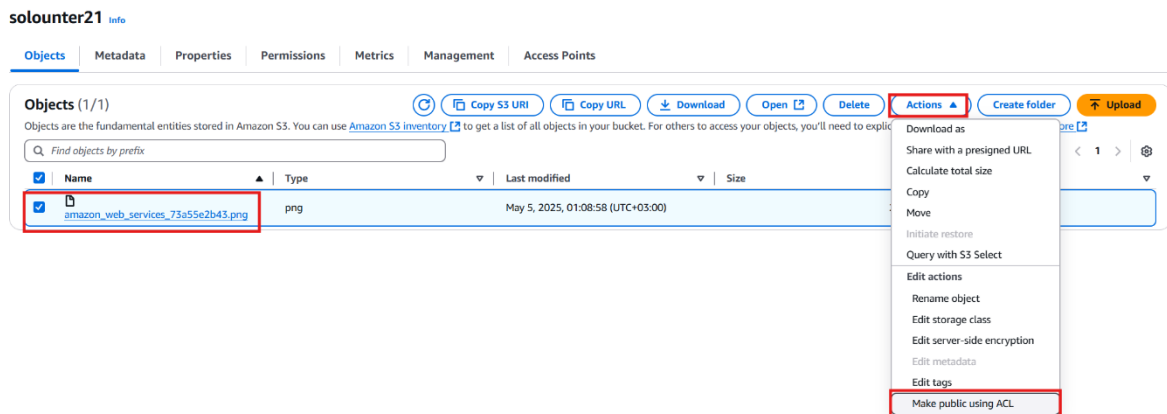
After uploading your file and folders click on **upload**.

This XML file does not appear to have any style information associated with it. The document tree is shown below.

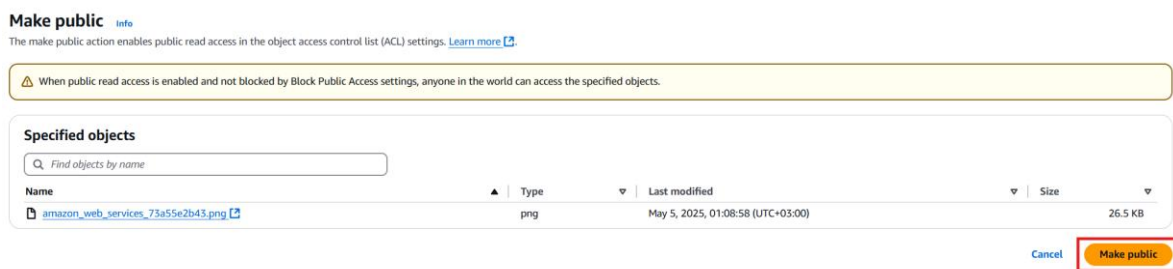
```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<Error>
  <Code>AccessDenied</Code>
  <Message>Access Denied</Message>
  <RequestId>FDX0SNFGZ28WQN9Z</RequestId>
  <HostId>DC+AHP292VcikoVKQ8pa0EkJO/1F8TbZ9UibJ+vavKENnRSJYvhuqTZQdY/26jN6mSEfSLP6Dx31IPkgKc4cWQy8nabjqg4h</HostId>
</Error>
  
```

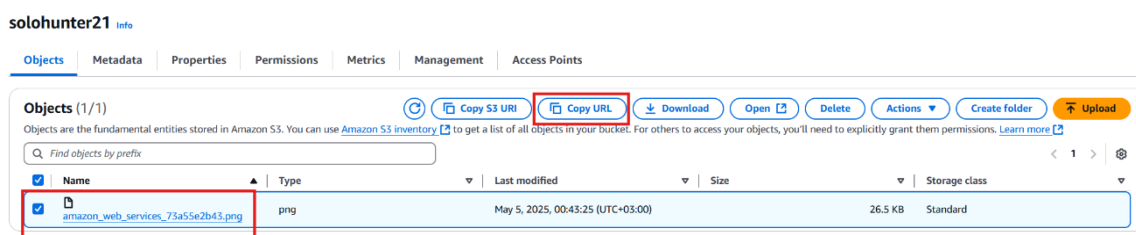
If you try to access the photo that you upload it will return **Access Denied** because of the **ACL** rules.



To give the photo permission to be accessed by the public, you select **Actions**, then **Make public using ACL**.



Click on **Make public**.



Select your photo the choose **copy URL** and passed it on **new tab**.

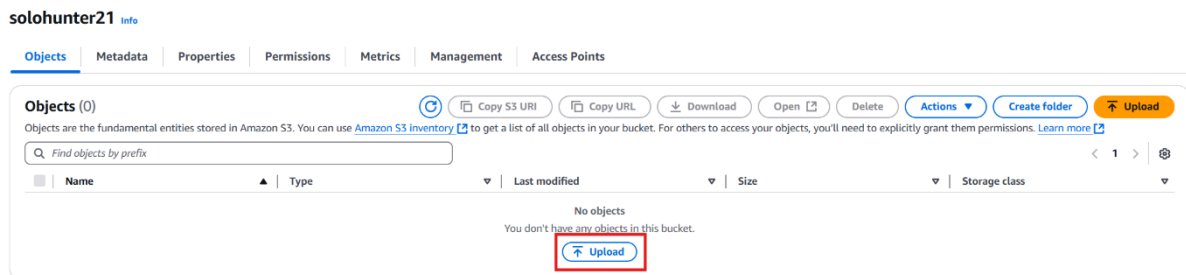


As you can see, the public can now access the photo **until you disable it**.

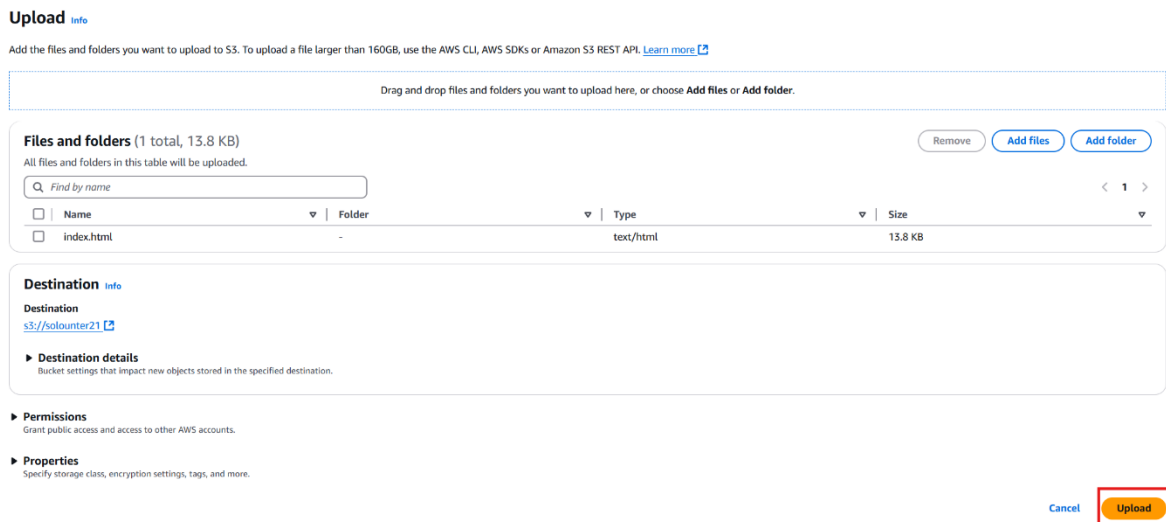
Create a Static Website in S3 Bucket

First, create a [bucket with ACL](#) enabled on it, same as in the above demonstration.

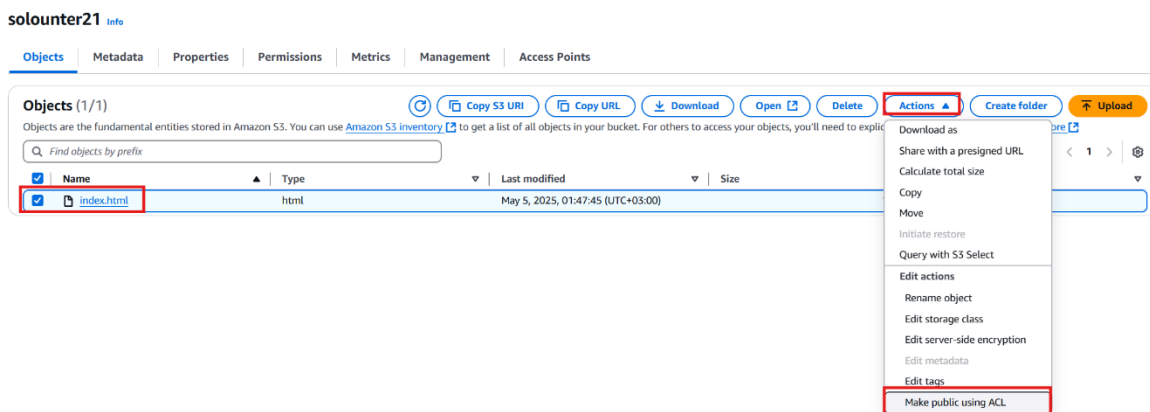
Second, you must have an **index.html** file ready.



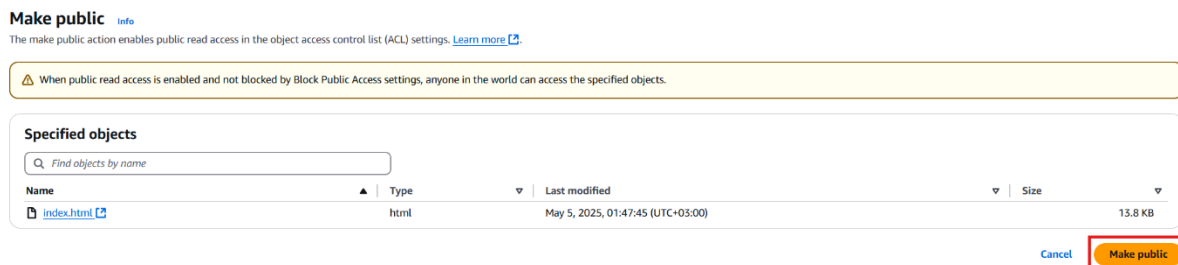
Click on **upload**.



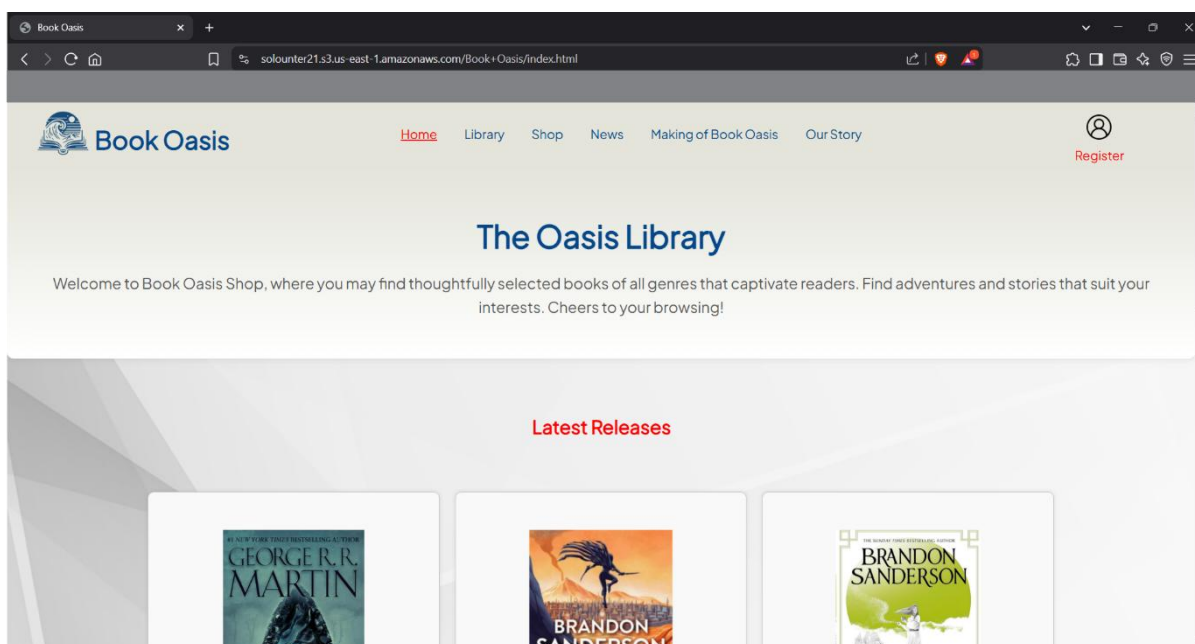
Select **Add file** and add your **index.html** file, then click on **upload**. Then return to your bucket.



Select your index file, click on Action, and make it public using ACL.



Choose make Public. Then go back to it and copy the URL. And paste in a new tab.



Now you can see the web page index.html is now accessible using s3 bucket.

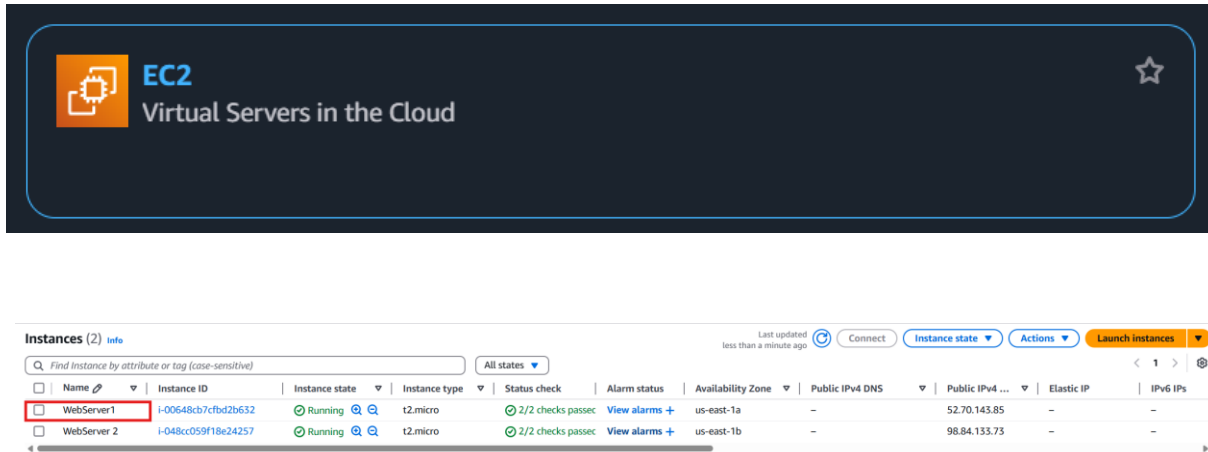
How to Create EBS Storage

Requirements:

1. [2 EC2](#) instances that run Ubuntu in different availability zones.
2. [2 public](#) subnets in different availability zones.

How to create and mount EBS storage in EC2 instance

First, EBS storage is suitable for high-performance workloads that require low latency. However, it must be mounted to a single EC2 instance and must exist in the same availability zone. Additionally, EBS volumes can only be expanded, not shrunk, and are supported only on **Linux**. A method to enable EBS functionality across different availability zones will be shown and explained below.



The screenshot shows the AWS Management Console interface for the EC2 console. The header includes the EC2 logo and the text "Virtual Servers in the Cloud". Below the header, there is a section titled "Instances (2)" with a search bar and a filter dropdown set to "All states". A table lists two EC2 instances:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
WebServer1	i-00648cb7cfbd2b632	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	52.70.143.85	-	-
WebServer 2	i-048cc059f18e24257	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	-	98.84.133.73	-	-

In EC2 instances, connect to one of your EC2 instances. In this demonstration, we will start with webserver 1.

- EC2
 - Dashboard
 - EC2 Global View 
 - Events
 - ▼ Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - ▼ Images
 - AMIs
 - AMI Catalog
 - ▼ Elastic Block Store
 - Volumes**
 - Snapshots
 - Lifecycle Manager

Volumes (2) [info](#)

Saved filter sets

Choose filter set ▾

Q Search

Last updated 1 minute ago

Actions ▾

Create volume

☐	Name ▾	Volume ID	Type ▾	Size ▾	IOPS ▾	Throughput ▾	Snapshot ID ▾	Created ▾	Availability Zone ▾	Volume state ▾	Alarm status ▾	Actions
☐	-	vol-08c70cd58ef1ad6e	gp3	8 GiB	3000	125	snap-0edeb0ff...	2025/05/06 17:37 GMT+3	us-east-1a	In-use	No alarms	+ info
☐	-	vol-00e450aa63182cc7	gp3	8 GiB	3000	125	snap-0edeb0ff...	2025/05/06 17:40 GMT+3	us-east-1b	In-use	No alarms	+ info

In the left-side menu, choose **Volumes**. You will notice that there are two volumes listed, which are associated with the EC2 instance you have created. Click on **Create Volume**.

Create volume

[Info](#)

Create an Amazon EBS volume to attach to any EC2 Instance in the same Availability Zone.

Volume settings

Volume type

[Info](#)

General Purpose SSD (gp3)

Size (GiB)

[Info](#)

100

Min: 1 GiB, Max: 16384 GiB.

IOPS

[Info](#)

3000

Min: 3000 IOPS, Max: 16000 IOPS.

Throughput (MiB/s)

[Info](#)

125

Min: 125 MiB, Max: 1000 MiB. Baseline: 125 MiB/s.

Availability Zone

[Info](#)

us-east-1a

Snapshot ID - optional

[Info](#)

Don't create volume from a snapshot



Encryption

[Info](#)

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☐ Encrypt this volume

Make sure you choose the **Availability zone** the same as your EC2 instance.

Tags - optional [Info](#)

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add tag](#)

You can add 50 more tags.

Snapshot summary [Info](#)

Click refresh to view backup information

The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

[Cancel](#)

[Create volume](#)

Leave the rest as the default options, and click on **Create Volume**.

Successfully created volume vol-0e6a49597d9f5da4e

Volumes (3) [Info](#)

Last updated less than a minute ago [Actions](#) [Create volume](#)

☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created	Availability Zone	Volume state	Alarm status	
☐	-	vol-08c70cf58ef1ad6e	gp3	8 GiB	3000	125	snap-0edbe0f...	2025/05/06 17:37 GMT+3	us-east-1a	In-use	No alarms	+ -
☐	-	vol-00e450aa63182cc7	gp3	8 GiB	3000	125	snap-0edbe0f...	2025/05/06 17:40 GMT+3	us-east-1b	In-use	No alarms	+ -
☐	-	vol-0e6a49597d9f5da4e	gp3	100 GiB	3000	125	-	2025/05/06 18:00 GMT+3	us-east-1a	Available	No alarms	+ -

Refresh the page; you will see that the volume has been created in volume state. You will notice that it says **Available**, which means it is **not attached to anything for the meantime**, and **give it a name**.

```
ubuntu@ip-192-168-1-233: ~$ sudo lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0        7:0      0   26.3M 1 loop /snap/amazon-ssm-agent/9881
loop1        7:1      0   73.9M 1 loop /snap/core22/1748
loop2        7:2      0   44.4M 1 loop /snap/snapd/23545
xvda        202:0     0    8G 0 disk
├─xvda1      202:1     0    7G 0 part /
├─xvda14     202:14    0    4M 0 part
├─xvda15     202:15    0   106M 0 part /boot/efi
└─xvda16     259:0     0   913M 0 part /boot
ubuntu@ip-192-168-1-233: ~$
```

Return to your EC2 and run the command “**sudo lsblk**” this command will list devices, partitions, mount points, and sizes.

Note: you can see all the devices that are connected to your Ubuntu because we did not mount the owner volume that we created, so it will not show.

Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID

vol-0e6a49597d9f5da4e (Volume 1)

Availability Zone

us-east-1a

Instance [Info](#)

Search instance ID or name tag

Q

i-00648cb7cfbd2b632 (WebServer1) (running)

[Cancel](#)
[Attach volume](#)

Go back to your AWS console and in volume, select your **volume** and click on **Action**, choose **Attach Volume**.

In the **Attach volume under the instance**, you will see only your instances that are in the same Availability zone. Choose your instances.

Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID

vol-0e6a49597d9f5da4e (Volume 1)

Availability Zone

us-east-1a

Instance [Info](#)

i-00648cb7cfbd2b632 (WebServer1) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)

/dev/sdk

Recommended device names for Linux: /dev/sda1 for root volume, /dev/sd[f-p] for data volumes.

[Cancel](#)
[Attach volume](#)

For the **Device Name**, you can choose any name, then click on **Attach Volume**.

Volumes (3) [Info](#)

Saved filter sets

Choose filter set

Q Search

Last updated 3 minutes ago

Actions

Create volume

☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created	Availability Zone	Volume state	Alarm status	At
☐	-	vol-08c7f0cf58ef1ad6e	gp3	8 GiB	3000	125	snap-0edbe0f...	2025/05/06 17:37 GMT+3	us-east-1a	in-use	No alarms	+ i-
☐	-	vol-00e450aa63182ccc7	gp3	8 GiB	3000	125	snap-0edbe0f...	2025/05/06 17:40 GMT+3	us-east-1b	in-use	No alarms	+ i-
☐	Volume 1	vol-0e6a49597d9f5da4e	gp3	100 GiB	3000	125	-	2025/05/06 18:00 GMT+3	us-east-1a	in-use	No alarms	+ i-

You will notice that the volume state changed to in-use and if you go again to your EC2

```
ubuntu@ip-192-168-1-233: ~$ sudo lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0        7:0      0  26.3M 1 loop /snap/amazon-ssm-agent/9881
loop1        7:1      0  73.9M 1 loop /snap/core22/1748
loop2        7:2      0  44.4M 1 loop /snap/snapd/23545
xvda        202:0      0    8G 0 disk
├─xvda1      202:1      0    7G 0 part /
├─xvda14     202:14     0    4M 0 part
├─xvda15     202:15     0  106M 0 part /boot/efi
├─xvda16     202:16     0  913M 0 part /boot
└─xvdk       202:160     0  100G 0 disk
```

instance and run the command again “**sudo lsblk**” you will notice it will show you a new volume.

```
ubuntu@ip-192-168-1-233: ~$ sudo lsblk
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop2: 44.44 MiB, 46596096 bytes, 91008 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/xvda: 8 GiB, 8589934592 bytes, 16777216 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: E3478E01-32E3-4FC2-8E79-1BCCDE89C2D7

Device        Start      End  Sectors  Size Type
/dev/xvda1    2099200 16777182 14677983   7G Linux filesystem
/dev/xvda14    2048    10239    8192    4M BIOS boot
/dev/xvda15   10240   227327   217088  106M EFI System
/dev/xvda16   227328 2097152 1869825  913M Linux extended boot

Partition table entries are not in disk order.

Disk /dev/xvdk: 100 GiB, 107374182400 bytes, 209715200 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
ubuntu@ip-192-168-1-233: ~$
```

In order to mount the drive to your Linux machine you has to have the name of the disk to get it run the command “**sudo fdisk -l**” this command will list all the disk in your Linux machine scroll down you will see the 100 gib disk that you attached early copy the name “/dev/xvdk” it could be different for you copy what is shown to you.

```
ubuntu@ip-192-168-1-233: ~  
ubuntu@ip-192-168-1-233:~$ sudo file -s /dev/xvdk  
/dev/xvdk: data  
ubuntu@ip-192-168-1-233:~$
```

If you run the command “**sudo file -s <your drive-name>**”, it will show you that it is a data that means it is raw, not formatted, and used by the Linux system.

```
ubuntu@ip-192-168-1-233: ~  
ubuntu@ip-192-168-1-233:~$ sudo mkfs -t xfs /dev/xvdk  
meta-data=/dev/xvdk          isize=512    agcount=4, agsize=6553600 blks  
=                               sectsz=512   attr=2, projid32bit=1  
=                               crc=1        finobt=1, sparse=1, rmapbt=1  
=                               reflink=1    bigtime=1 inobtcount=1 nrext64=0  
data      =                   bsize=4096   blocks=26214400, imaxpct=25  
=                               sunit=0      swidth=0 blks  
naming    =version 2          bsize=4096   ascii-ci=0, ftype=1  
log       =internal log      bsize=4096   blocks=16384, version=2  
=                               sectsz=512   sunit=0 blks, lazy-count=1  
realtime  =none              extsz=4096   blocks=0, rtextents=0  
ubuntu@ip-192-168-1-233:~$
```

To use the drive, you must make a file system on it by running the command “**sudo mkfs -t <file system> <name of the drive>**”

This command will create a file system on the drive so you can use it.

```
ubuntu@ip-192-168-1-233: ~  
ubuntu@ip-192-168-1-233:~$ sudo file -s /dev/xvdk  
/dev/xvdk: SGI XFS filesystem data (blksz 4096, inosz 512, v2 dirs)  
ubuntu@ip-192-168-1-233:~$
```

To check if the drive creates the file system by running the command “**sudo file -s <drive name>**”, as you can see, the file system has been created on the drive.

```
ubuntu@ip-192-168-1-233: ~  
ubuntu@ip-192-168-1-233:~$ sudo mkdir /EBSstorage
```

To mount the drive, you must have a folder to mount it on. First, create a Folder by running the command “**sudo mkdir <directory name>**”

```
ubuntu@ip-192-168-1-233: ~  
ubuntu@ip-192-168-1-233:~$ sudo mount /dev/xvdk /EBSstorage/
```

After creating the directory, you must mount the drive on the directory by issuing the command “**sudo mount <drive name> <directory location and name>**”.

```
ubuntu@ip-192-168-1-233: /EBSstorage/  
ubuntu@ip-192-168-1-233:~$ cd /EBSstorage/  
ubuntu@ip-192-168-1-233:/EBSstorage$ sudo touch test.txt  
ubuntu@ip-192-168-1-233:/EBSstorage$ ls  
test.txt  
ubuntu@ip-192-168-1-233:/EBSstorage$
```

After mounting the drive, go to the directory and **create a file in it**.

Create Snapshot from EBS

EC2

Dashboard

EC2 Global View [↗](#)

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Snapshots (1) [info](#)

Owned by me [▼](#)

Last updated
less than a minute ago

[Recycle Bin](#)

[Actions](#)

Create snapshot

☐	Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot status	Started	Progress	Encryption
☐	-	snap-06644a08ee0d09289	8 GiB	8 GiB	Created by CreateImagef(...)	Standard	Completed	2025/04/23 15:47 GMT+3	100%	Not encrypted

Return to your AWS console and in the left-side menu in your **EC2** select **snapshot**, then click on the **snapshot**.

Note: because of the limitation of the EBS, you cannot mount EBS in two different EC2 instances in different Availability zones, but you can take a copy of the drive by using a snapshot.

Create snapshot [info](#)

Create a point-in-time snapshot of an EBS volume and use it as a baseline for new volumes or for data backup. You can create snapshots from an individual volume, or you can create multi-volume snapshots from all of the volumes attached to an instance.

Source

Resource type [info](#)

☒ **Volume**
Create a snapshot from a specific volume.

☐ **Instance**
Create multi-volume snapshots from an instance.

Volume ID
The volume from which to create the snapshot.

vol-0e6a49597d9f5da4e (Volume 1)

Snapshot details

Description
Add a description for your snapshot.

255 characters maximum

Encryption [info](#)
Not encrypted

Tags [info](#)
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.

[Add tag](#)
You can add 50 more tags.

[Cancel](#)
[Create snapshot](#)

First, make sure you select the Volume for the Resource type. Second, select your Volume. Third, write a meaningful description. Last click on **create snapshot**.

Snapshots (2) [info](#)

Owned by me

	Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot status	Started	Progress	Encryption
☐	-	snap-096632afac1eb3973	-	100 GiB	this is a snapshot form web...	Standard	Pending	2025/05/06 19:07 GMT+3	0%	Not encrypted
☐	-	snap-06644a08ee0d09289	8 GiB	8 GiB	Created by CreatelImageI-...	Standard	Completed	2025/04/23 15:47 GMT+3	100%	Not encrypted

After creating the snapshot, wait for some time for the snapshot state to say **complete**.

Snapshots (1/2) [info](#)

Owned by me

	Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot status	Started	Progress	Encryption
☒	-	snap-096632afac1eb3973	66.5 MiB	100 GiB	this is a snapshot form web...	Standard	Completed	2025/05/06		Not encrypted
☐	-	snap-06644a08ee0d09289	8 GiB	8 GiB	Created by CreatelImageI-...	Standard	Completed	2025/04/23		Not encrypted

[Recycle Bin](#)
[Actions](#)

[Create volume from snapshot](#)
[Create image from snapshot](#)
[Copy snapshot](#)
[Launch copy duration calculator](#)
[Delete snapshot](#)
[Manage tags](#)
[Snapshot settings](#)
[Archiving](#)

After the snapshot is completed, select the ID and click on **Action**, choose **Create Volume from snapshot**.

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Snapshot ID
[snap-096632afac1eb3973](#)

Volume type [Info](#)
 General Purpose SSD (gp3)

Size (GiB) [Info](#)
 100
Min: 1 GiB, Max: 16384 GiB.

IOPS [Info](#)
 3000
Min: 3000 IOPS, Max: 16000 IOPS.

Throughput (MiB/s) [Info](#)
 125
Min: 125 MiB, Max: 1000 MiB, Baseline: 125 MiB/s.

Availability Zone [Info](#)
 us-east-1b

Fast snapshot restore [Info](#)
 Not enabled for selected snapshot

Encryption
 Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.
☐ Encrypt this volume

Tags - optional [Info](#)
 A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
 No tags associated with the resource.
[Add tag](#)
 You can add 50 more tags.

Snapshot summary [Info](#)
 Click refresh to view backup information
 The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

[Cancel](#) [Create volume](#)

Select your **Availability Zone** that matches your **second EC2 instance**. Leave the rest as a default and click on Create Volume.

Volumes (4) [Info](#)

Last updated less than a minute ago [Actions](#) [Create volume](#)

☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created	Availability Zone	Volume state	Alarm status	At
☐	-	vol-08c7f0cf58ef1ad6e	gp3	8 GiB	3000	125	snap-0edbe0f...	2025/05/06 17:37 GMT+3	us-east-1a	In-use	No alarms	+ i-
☐	-	vol-00e450aa63182ccc7	gp3	8 GiB	3000	125	snap-0edbe0f...	2025/05/06 17:40 GMT+3	us-east-1b	In-use	No alarms	+ i-
☐	-	vol-0942e768e0881343e	gp3	100 GiB	3000	125	snap-096632a...	2025/05/06 19:14 GMT+3	us-east-1b	Available	No alarms	+ -
☐	Volume 1	vol-0e6a49597d9f5da4e	gp3	100 GiB	3000	125	-	2025/05/06 18:00 GMT+3	us-east-1a	In-use	No alarms	+ i-

From the left-side menu, **select volume**. You will find the new volume **that you created from the snapshot**. Give it a name.

Volumes (1/4) [Info](#)

Last updated 2 minutes ago [Actions](#) [Create volume](#)

☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created	Availability Zone	Volume state	Alarm status	At
☐	-	vol-08c7f0cf58ef1ad6e	gp3	8 GiB	3000	125	snap-0edbe0f...	2025/05/06 17:37 GMT+3	us-east-1a	In-use	No alarms	+ i-
☐	-	vol-00e450aa63182ccc7	gp3	8 GiB	3000	125	snap-0edbe0f...	2025/05/06 17:40 GMT+3	us-east-1b	In-use	No alarms	+ i-
☒	Volume 2	vol-0942e768e0881343e	gp3	100 GiB	3000	125	snap-096632a...	2025/05/06 19:14 GMT+3	us-east-1b	Available	No alarms	+ -
☐	Volume 1	vol-0e6a49597d9f5da4e	gp3	100 GiB	3000	125	-	2025/05/06 18:00 GMT+3	us-east-1a	In-use	No alarms	+ i-

[Modify volume](#)
[Create snapshot](#)
[Create snapshot lifecycle policy](#)
[Delete volume](#)
[Attach volume](#)
[Detach volume](#)
[Force detach volume](#)
[Manage auto-enabled I/O](#)
[Manage tags](#)
[Fault injection](#)

Now, you must **attach the Volume to your second EC2 instance**. Select the volume you want to attach, click on **Action**, and choose from the drop-down menu **Attach volume**.

Attach volume [info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID

vol-0942e768e0881343e (Volume 2)

Availability Zone

us-east-1b

Instance

i-048cc059f18e24257

(WebServer 2) (running)

Device name

/dev/sdk

Newer Linux kernels may rename your devices to `/dev/xvdf` through `/dev/xvdp` internally, even when the device name entered here (and shown in the details) is `/dev/sdf` through `/dev/sdp`.

Cancel

Attach volume

Select your **second EC2 instance** and choose a Device name, then click on **Attach volume**.

```

ubuntu@ip-192-168-2-210: ~$ sudo lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0         7:0    0  26.3M  1 loop /snap/amazon-ssm-agent/9881
loop1         7:1    0  73.9M  1 loop /snap/core22/1748
loop2         7:2    0  44.4M  1 loop /snap/snapd/23545
xvda         202:0    0    8G  0 disk
├─xvda1       202:1    0    7G  0 part /
├─xvda14      202:14   0    4M  0 part 
├─xvda15      202:15   0   106M  0 part /boot/efi
└─xvda16      202:16   0   913M  0 part /boot
xvdk         202:160  0  100G  0 disk
ubuntu@ip-192-168-2-210: ~$
  
```

Connect to your second EC2 instance and update Ubuntu, then check if the disk is attached, run the command “**sudo lsblk**”; as you can see, the disk is attached.

```
ubuntu@ip-192-168-2-210: ~$ sudo fdisk -l

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop2: 44.44 MiB, 46596096 bytes, 91008 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/xvda: 8 GiB, 8589934592 bytes, 16777216 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: E3478E01-32E3-4FC2-8E79-1BCCDE89C2D7

Device            Start      End  Sectors  Size Type
/dev/xvda1        2099200    16777182 14677983   7G Linux filesystem
/dev/xvda14         2048      10239    8192     4M BIOS boot
/dev/xvda15        10240     227327   217088   106M EFI System
/dev/xvda16       227328    2097152  1869825   913M Linux extended boot

Partition table entries are not in disk order.

Disk /dev/xvdk: 100 GiB, 107374182400 bytes, 209715200 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
ubuntu@ip-192-168-2-210:~$
```

To check the file system of the drive. First, you must know the name of the drive by running the command “**sudo fdisk -l**” to list all the drives, and scroll down to see your disk name. Copy it.

```
ubuntu@ip-192-168-2-210: ~$ sudo fdisk -l

Disk /dev/loop2: 44.44 MiB, 46596096 bytes, 91008 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/xvda: 8 GiB, 8589934592 bytes, 16777216 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: E3478E01-32E3-4FC2-8E79-1BCCDE89C2D7

Device            Start      End  Sectors  Size Type
/dev/xvda1        2099200    16777182 14677983   7G Linux filesystem
/dev/xvda14         2048      10239    8192     4M BIOS boot
/dev/xvda15        10240     227327   217088   106M EFI System
/dev/xvda16       227328    2097152  1869825   913M Linux extended boot

Partition table entries are not in disk order.

Disk /dev/xvdk: 100 GiB, 107374182400 bytes, 209715200 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
ubuntu@ip-192-168-2-210:~$ sudo file -s /dev/xvdk
/dev/xvdk: SGI XFS filesystem data (blkisz 4096, inosz 512, v2 dirs)
ubuntu@ip-192-168-2-210:~$
```

After obtaining the name of the drive, you can check the file system of it by issuing the command “**sudo file -s <drive name>**”. As you can see, the drive is using the XFS file system.

```

ubuntu@ip-192-168-2-210: /Ei x + v

Disk /dev/xvda: 8 GiB, 8589934592 bytes, 16777216 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: E3478E01-32E3-4FC2-8E79-1BCCDE89C2D7

Device      Start      End      Sectors  Size Type
/dev/xvda1  2099200    16777182 14677983  7G Linux filesystem
/dev/xvda14  2048       10239     8192     4M BIOS boot
/dev/xvda15  10240      227327   217088   106M EFI System
/dev/xvda16  227328     2097152  1869825  913M Linux extended boot

Partition table entries are not in disk order.

Disk /dev/xvdk: 100 GiB, 107374182400 bytes, 209715200 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
ubuntu@ip-192-168-2-210:~$ sudo file -s /dev/xvdk
/dev/xvdk: SGI XFS filesystem data (blkisz 4096, inosz 512, v2 dirs)
ubuntu@ip-192-168-2-210:~$ sudo mkdir /EBS-storage
ubuntu@ip-192-168-2-210:~$ sudo mount /dev/xvdk /EBS-storage/
ubuntu@ip-192-168-2-210:~$ cd /EBS-storage/
ubuntu@ip-192-168-2-210:/EBS-storage$ ls
test.txt
ubuntu@ip-192-168-2-210:/EBS-storage$

```

Now you need to mount the drive to a directory. First make a directory by using the command “**sudo mkdir <directory name>**” Second, mount the drive by running the command “**sudo mount <drive name> <directory location and name>**” Third, to check If the file you created in the first instance is there or not issue the command “**ls**” will list everything in the folder and that is how to create a copy of your EBS drive and mounted in different EC2 instance in different availability zone.

If you have done everything, the correct file will be there.

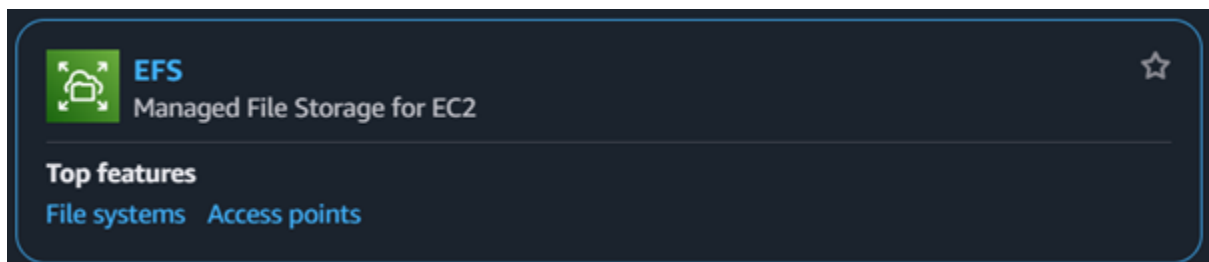
How to Create EFS Storage

Requirements:

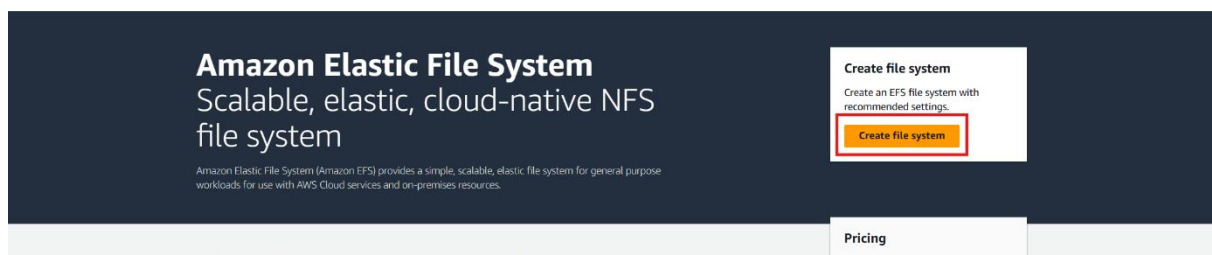
3. [2 EC2](#) instances that run Ubuntu in different availability zones.
4. [2 public](#) subnets in different availability zones.

How to create and mount EFS Storage in EC2 instance

EFS Storage can be mounted in multiple EC2 instances, and at the same time, you can shrink or expand your drive because it is elastic. It is best if you need shared storage across multiple instances. In this demonstration, we will use a single EFS volume to be connected to 2 EC2 instances in different Availability zones.



Create EFS File System



Go to **EFS** and click on **Create file system**.

Create file system



Create a file system with the recommended settings shown below by choosing Create file system. To view all settings or to customize your file system, choose Customize. [Learn more](#)

Name - optional

Name your file system.

Optional. Apply a name to your file system

Name can include letters, numbers, and +-=._:/ symbols, up to 256 characters.

Virtual Private Cloud (VPC)

Choose the VPC where you want EC2 instances to connect to your file system.

vpc-0e59dea41fb14b6b2

Hussain-VPC

Recommended settings

Your file system is created with the following recommended settings unless you choose to customize the file system. You will be charged for storage and throughput. We recommend reviewing pricing for these features using the [AWS Pricing Calculator](#)

Setting	Value	Editable after creation
Throughput mode Learn more	Elastic	Yes
Transition into Infrequent Access (IA)	30 day(s) since last access	Yes
Transition into Archive	90 day(s) since last access	Yes
Transition into Standard	None	Yes
Automatic backups	Enabled	Yes
Encryption	Enabled	No

Cancel

Customize

Create file system

Choose a **name** for your file system and **choose your VPC**, then click on create file system.

Amazon EFS > File systems

File systems (1)

Filter by property values

	Name	File system ID	Encryption	Total size	Size in Standard	Size in IA	Size in Archive	Provisioned Throughput (MiB/s)	File system state	Creation time	Availability Zone	Replication overwrite protection
☐	Hussain-Filesystem	fs-01f39dff547096b1c	Encrypted	6.00 KiB	6.00 KiB	0 Bytes	0 Bytes	-	Available	Tue, 06 May 2025 16:53:18 GMT	Regional	Enabled

After successfully creating the file system, click on your file system.

Hussain-Filesystem (fs-01f39dff547096b1c) Delete Attach

General Edit

Amazon resource name (ARN)
arn:aws:elasticfilesystem:us-east-1:533267121457:file-system/fs-01f39dff547096b1c

Performance mode
General Purpose

Throughput mode
Elastic

Lifecycle management
Transition into Infrequent Access (IA): 30 day(s) since last access
Transition into Archive: 90 day(s) since last access
Transition into Standard: None

Availability zone
Regional

Automatic backups
Enabled

Encrypted
8367d016-5654-4391-92f0-70b82d1d3822 (aws/elasticfilesystem)

File system state
Available

DNS name
fs-01f39dff547096b1c.efs.us-east-1.amazonaws.com

Replication overwrite protection
Enabled

Metered size | Monitoring | Tags | File system policy | Access points | **Network** | Replication

Network Refresh Manage

Availability zone (AZ-ID)	Mount target ID	Subnet ID	Mount target state	IP address	Network interface ID	Security groups
us-east-1a (use1-az2)	fsmt-05f2314a85938977c	subnet-0743a9104c2178479	Creating	192.168.1.235	eni-052314e96c27998a5	-
us-east-1b (use1-az4)	fsmt-04aa38a484e1427d4	subnet-0f5f2d70b67028b18	Creating	192.168.2.125	eni-0af56479a58bbba18	-

Scroll down and choose from the drop-down menu, **Network**, you will notice it is creating for you in every Availability zone your VPC has. After the mount target becomes Available, click on **Manage**.

Network

Virtual Private Cloud (VPC)
Choose the VPC where you want EC2 instances to connect to your file system.
vpc-0e59dea41fb14b6b2
Hussain-VPC
You must delete all existing mount targets in order to change the VPC of your file system.

Mount targets
A mount target provides an NFSv4 endpoint at which you can mount an Amazon EFS file system. We recommend creating one mount target per Availability Zone. [Learn more](#)

Availability zone	Subnet ID	IP address	Security groups
us-east-1a	subnet-0743a9104c2178479	192.168.1.235	Choose security groups sg-0f1c3c936a51c50c9 X default Remove
us-east-1b	subnet-0f5f2d70b67028b18	192.168.2.125	Choose security groups sg-0f1c3c936a51c50c9 X default Remove

Add mount target

Cancel Save

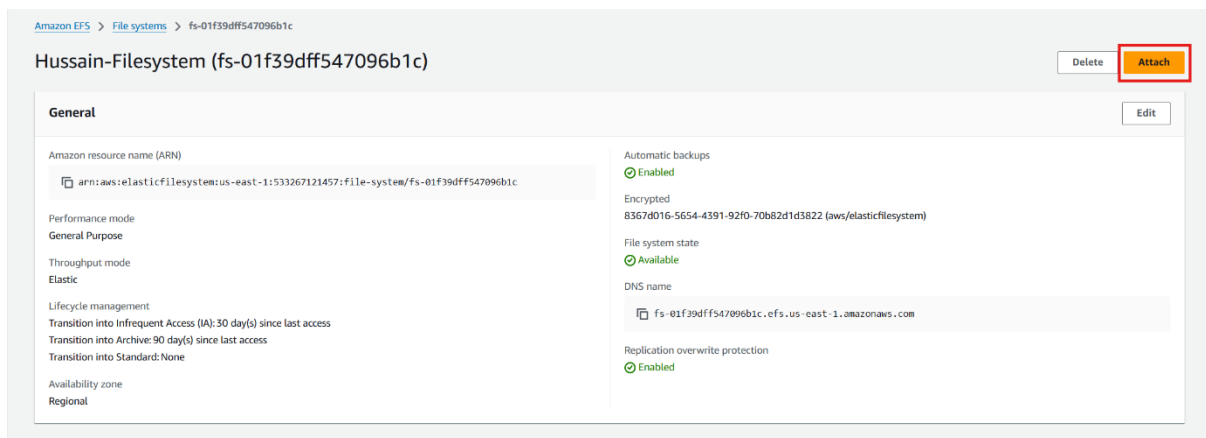
In this tab, you can remove the filesystem that is in the Availability zone you don't need, and remove the default option of the security group from your security group that you created before, then click save.

Note: In some cases, if you click on save, it will give you an error message saying failed. All you have to do is click back on manage, make sure that you did everything correctly, and click on save again, and it will *hopefully* give you a success message.

Mounting the EFS drive to EC2 instance

```
ubuntu@ip-192-168-1-99: ~$ sudo apt-get install nfs-common
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  keyutils libnfsidmap1 rpcbind
Suggested packages:
  watchdog
The following NEW packages will be installed:
  keyutils libnfsidmap1 nfs-common rpcbind
0 upgraded, 4 newly installed, 0 to remove and 82 not upgraded.
Need to get 400 kB of archives.
After this operation, 1416 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Connect to your EC2 instance and update Ubuntu, then install the nfs client to mount the EFS volume by running the command “**sudo apt-get install nfs-common**”. You will be asked to continue, enter “**Y**”.



Amazon EFS > File systems > fs-01f39dff547096b1c

Hussain-Filesystem (fs-01f39dff547096b1c)

[Delete](#) [Attach](#)

General [Edit](#)

Amazon resource name (ARN)
arn:aws:elasticfilesystem:us-east-1:533267121457:file-system/fs-01f39dff547096b1c

Performance mode
General Purpose

Throughput mode
Elastic

Lifecycle management
Transition into Infrequent Access (IA): 30 day(s) since last access
Transition into Archive: 90 day(s) since last access
Transition into Standard: None

Availability zone
Regional

Automatic backups
Enabled

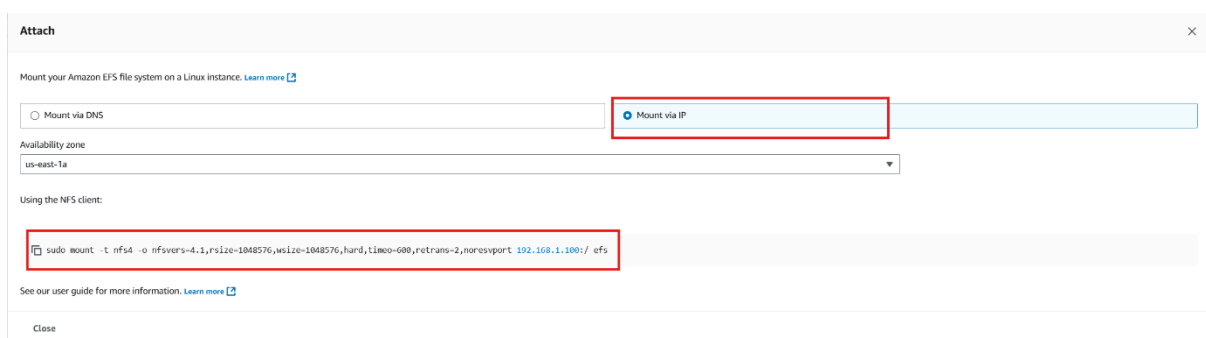
Encrypted
8367d016-5654-4391-92f0-70b82d1d3822 (aws/elasticfilesystem)

File system state
Available

DNS name
fs-01f39dff547096b1c.efs.us-east-1.amazonaws.com

Replication overwrite protection
Enabled

Go back to your file system and click on Attach.



Attach

Mount your Amazon EFS file system on a Linux instance. [Learn more](#)

☐ Mount via DNS ☒ Mount via IP

Availability zone
us-east-1a

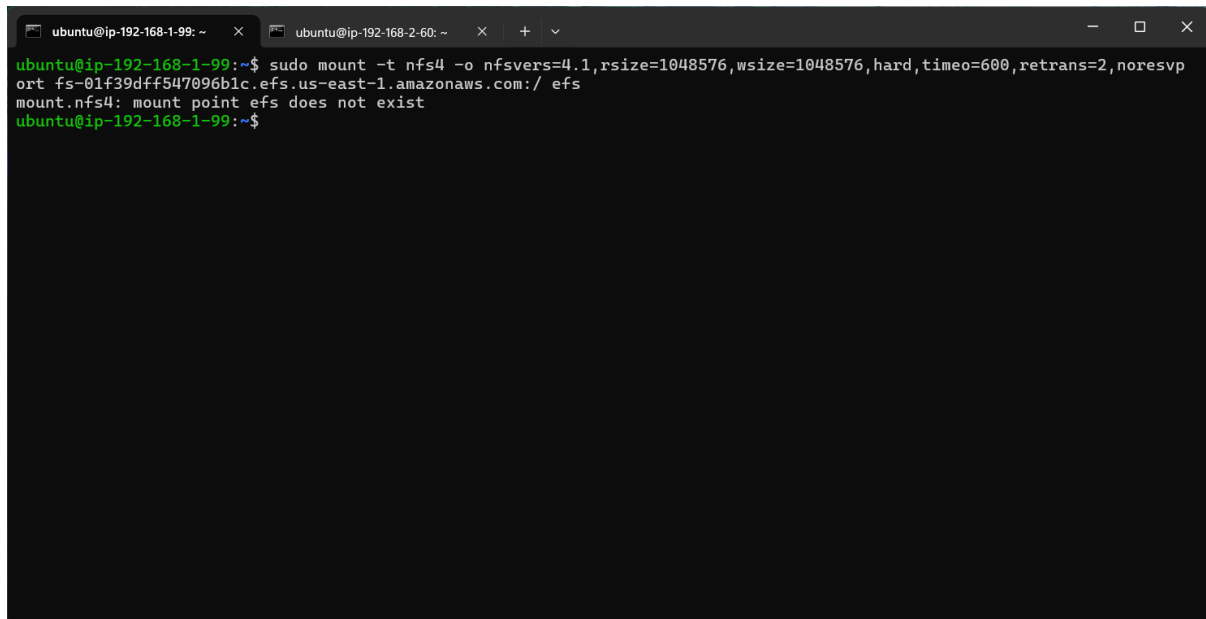
Using the NFS client:

```
sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsiz=1048576,hard,timeo=600,retrans=2,noresvport 192.168.1.100:/ efs
```

See our user guide for more information. [Learn more](#)

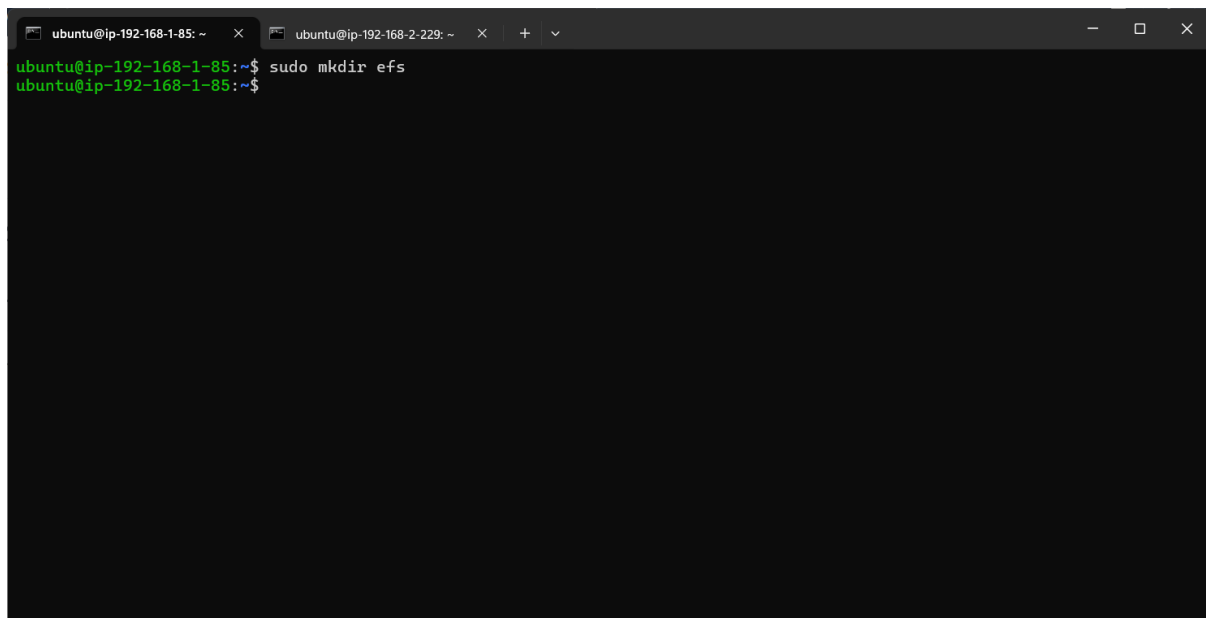
Close

Choose **mount via IP** and **select your availability zone for your first EC2 instance**, then copy the command.



```
ubuntu@ip-192-168-1-99: ~  
ubuntu@ip-192-168-2-60: ~  
ubuntu@ip-192-168-1-99:~$ sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsiz=1048576,hard,timeo=600,retrans=2,noresvport fs-01f39dff547096b1c.efs.us-east-1.amazonaws.com:/ efs  
mount.nfs4: mount point efs does not exist  
ubuntu@ip-192-168-1-99:~$
```

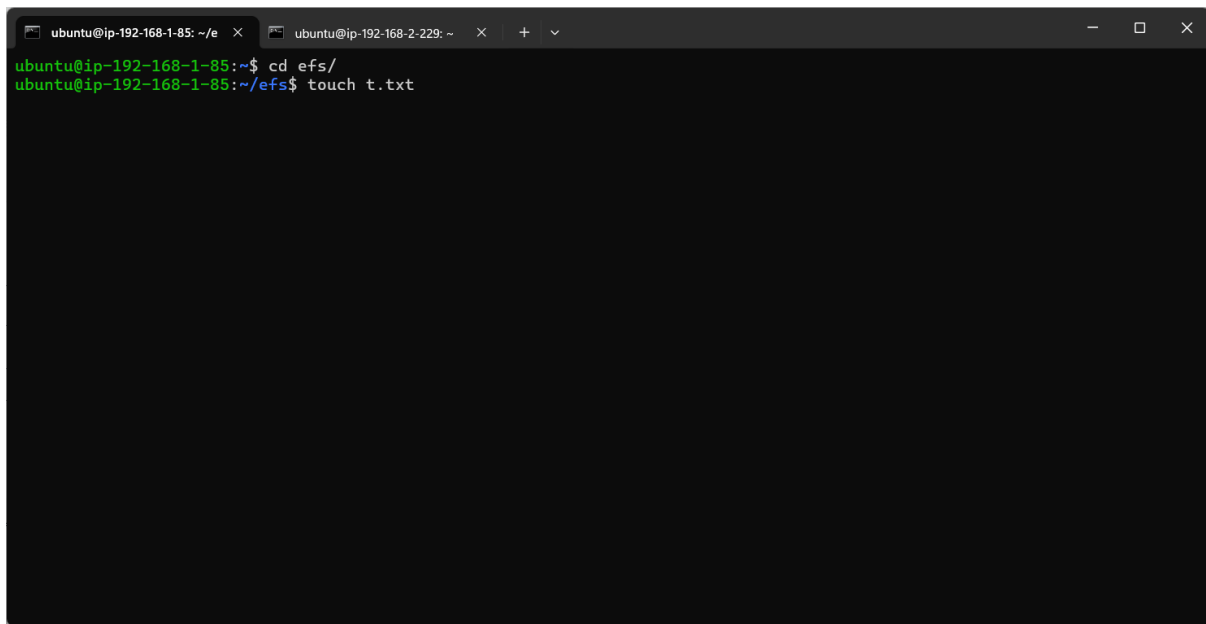
If you try to run the command, you will get an error message that says **mount point efs does not exist**. The reason for the error message is that there is **no efs directory** to mount the drive on it. As you know, you can't mount a drive without specifying the directory in Linux.



```
ubuntu@ip-192-168-1-85: ~  
ubuntu@ip-192-168-2-229: ~  
ubuntu@ip-192-168-1-85:~$ sudo mkdir efs  
ubuntu@ip-192-168-1-85:~$
```

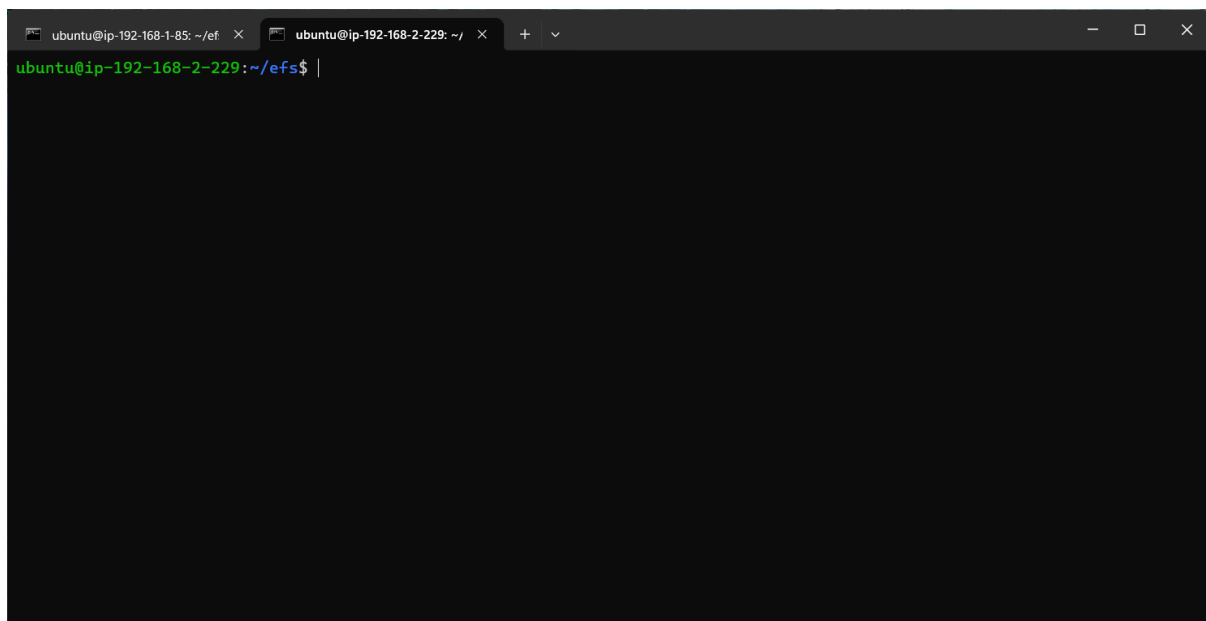
You must create a directory named **“efs”** and then paste the command you copied.

Mounting EFS drive in Different Availability zone

A terminal window with two tabs. The first tab is titled "ubuntu@ip-192-168-1-85: ~/ef" and the second is "ubuntu@ip-192-168-2-229: ~". The first tab shows the commands "cd efs/" and "touch t.txt" being executed. The second tab is currently active and shows a blank prompt.

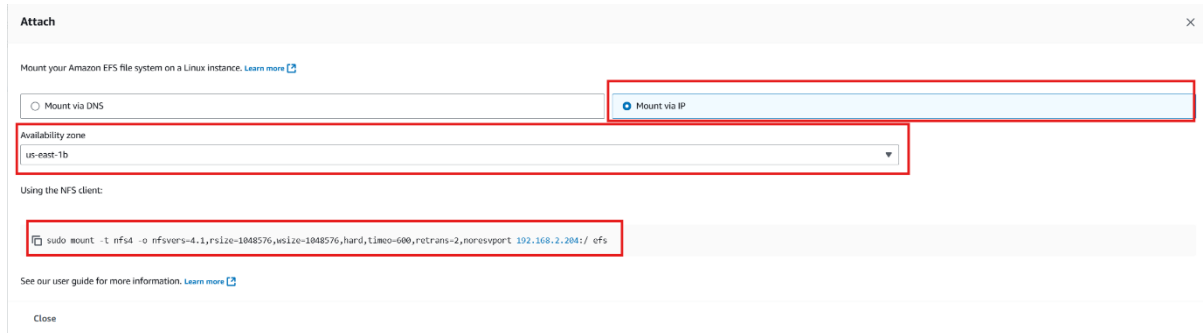
```
ubuntu@ip-192-168-1-85:~$ cd efs/  
ubuntu@ip-192-168-1-85:~/efs$ touch t.txt
```

Now, go into your EFS directory and create a file to test if the drive is working in multiple Availability zones.

A terminal window with two tabs. The first tab is titled "ubuntu@ip-192-168-1-85: ~/ef" and the second is "ubuntu@ip-192-168-2-229: ~/efs". The second tab is active and shows a blank prompt.

```
ubuntu@ip-192-168-2-229:~/efs$ |
```

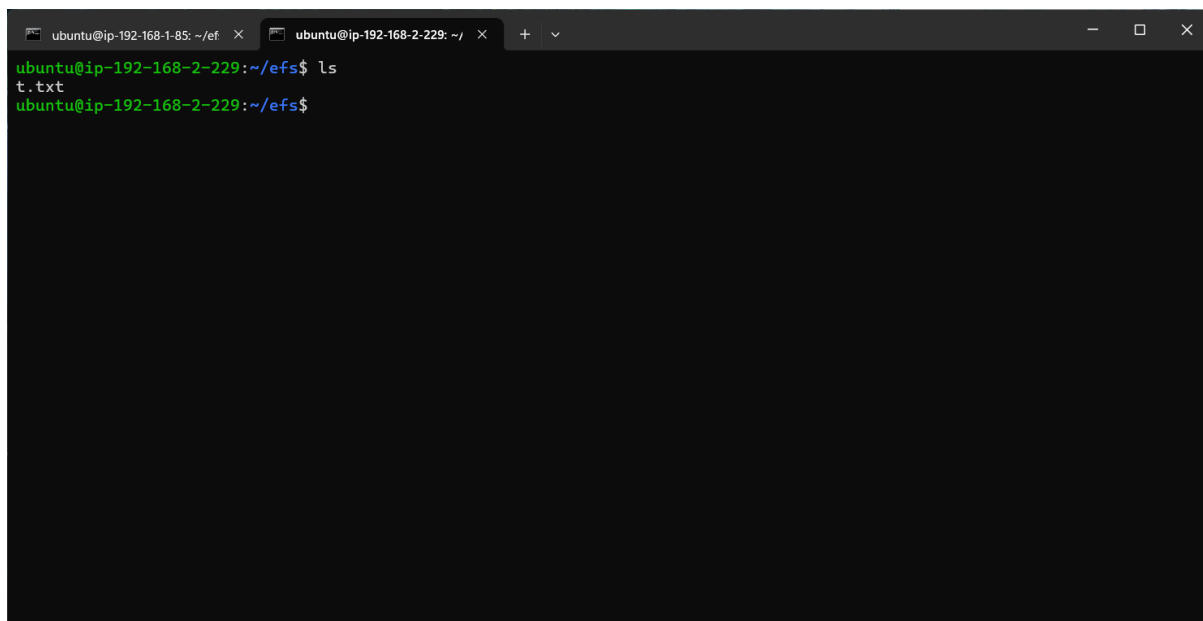
Now **connect to your second EC2 instance** and **update Ubuntu** and **install the nfs client** by running the command “**sudo apt-get install nfs-common**”.



The 'Attach' dialog box shows options for mounting an Amazon EFS file system. The 'Mount via IP' option is selected. The 'Availability zone' dropdown is set to 'us-east-1b'. The 'Using the NFS client' section contains a terminal command: `sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsz=1048576,hard,timeo=600,retrans=2,noresvport 192.168.2.204:/ efs`.

Go back to your **EFS file system** and click on **Attach**, choose **mount via IP**, and choose your **second availability zone** where your **second EC2 instance** is, **copy the command** and **paste in your EC2 instance** after you create a **directory named efs** by running the command “**sudo mkdir efs**”

Verifying Cross-Instance File Availability Between Two EC2 Instances



The terminal window shows the command `ls` being executed in the `~/efs` directory. The output is `t.txt`, indicating that the file created in the first EC2 instance is accessible from the second EC2 instance.

Go inside your EFS directory and run “**ls**” to list the contents of the file. You will see the file you created in the first EC2 instance.

Note: the benefit of EFS is that it can be mounted in different Availability zones and have the same data without the fuss of creating a snapshot and mount the snapshot in EBS

GOOD LUCK